Phys 153: Fundamentals of Physics III

Unit #4 – Old Quantum Theory

Aaron Wirthwein

wirthwei@usc.edu | SHS 370

Department of Physics and Astronomy

University of Southern California

Textbook Sections

Please read the following sections carefully:

- Ch 37.6
- Ch 39 (all of it)
- Ch 41.1-41.3

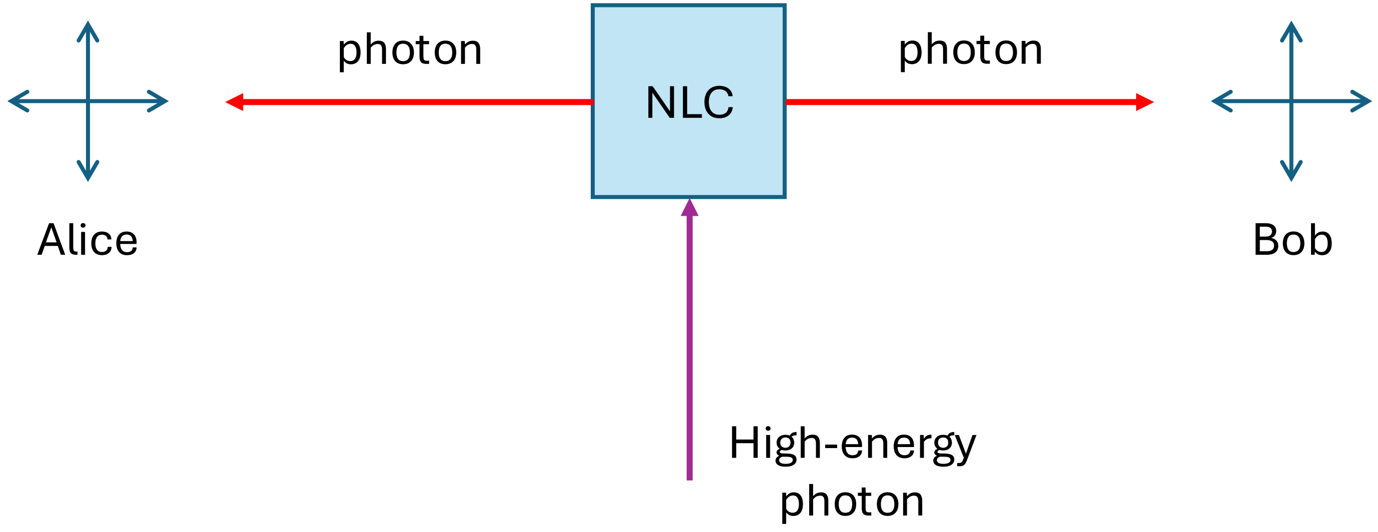
As in the unit on special relativity, we will heavily supplement the textbook in lecture.

Entanglement

- Last lecture, we discussed single-particle interference as a kind of interference that cannot be explained by any classical principles (quantum-mechanical interference)
  - See Class Notes 03-26-26
- Another prime example of non-classical behavior that requires a fully quantum description is entanglement.
  - Informally, entanglement happens when the state of a system cannot be neatly factored into independent parts.

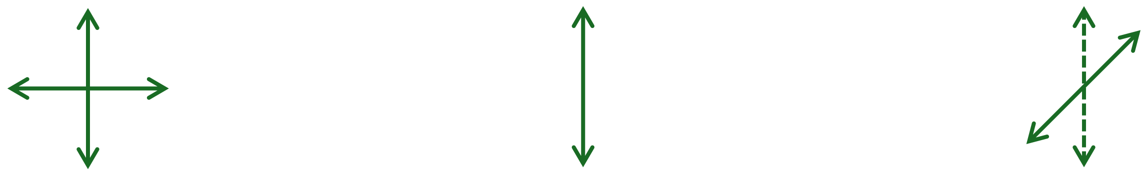
Entangled Photons

- Entangled photons can be created by an optical process called spontaneous parametric down conversion.
  - Laser photon passes through a nonlinear crystal, a material that interacts with light in a way that produces a change in frequency, phase, polarization, amplitude, etc.
  - Photon "splits" into two lower-energy photons and conservation laws (energy, momentum, angular momentum) force the pair into a correlated state.
  - Under the right circumstances, state could be entangled!



Polarization of Light Waves

- Classically, polarization describes the direction in which the electric field of an EM wave oscillates.
- Light composed of many EM waves polarized in all directions perpendicular to line of travel is called unpolarized. By superposition, no net polarization.
- Since EM waves are transverse, there are only two linearly independent polarization states.

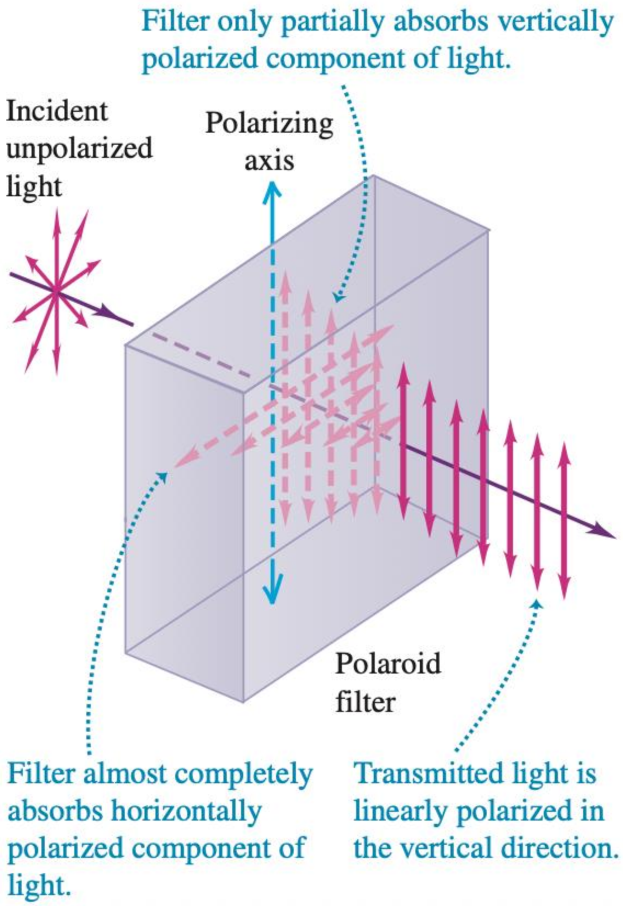


unpolarized linear (H/V) circular (CW/CCW)

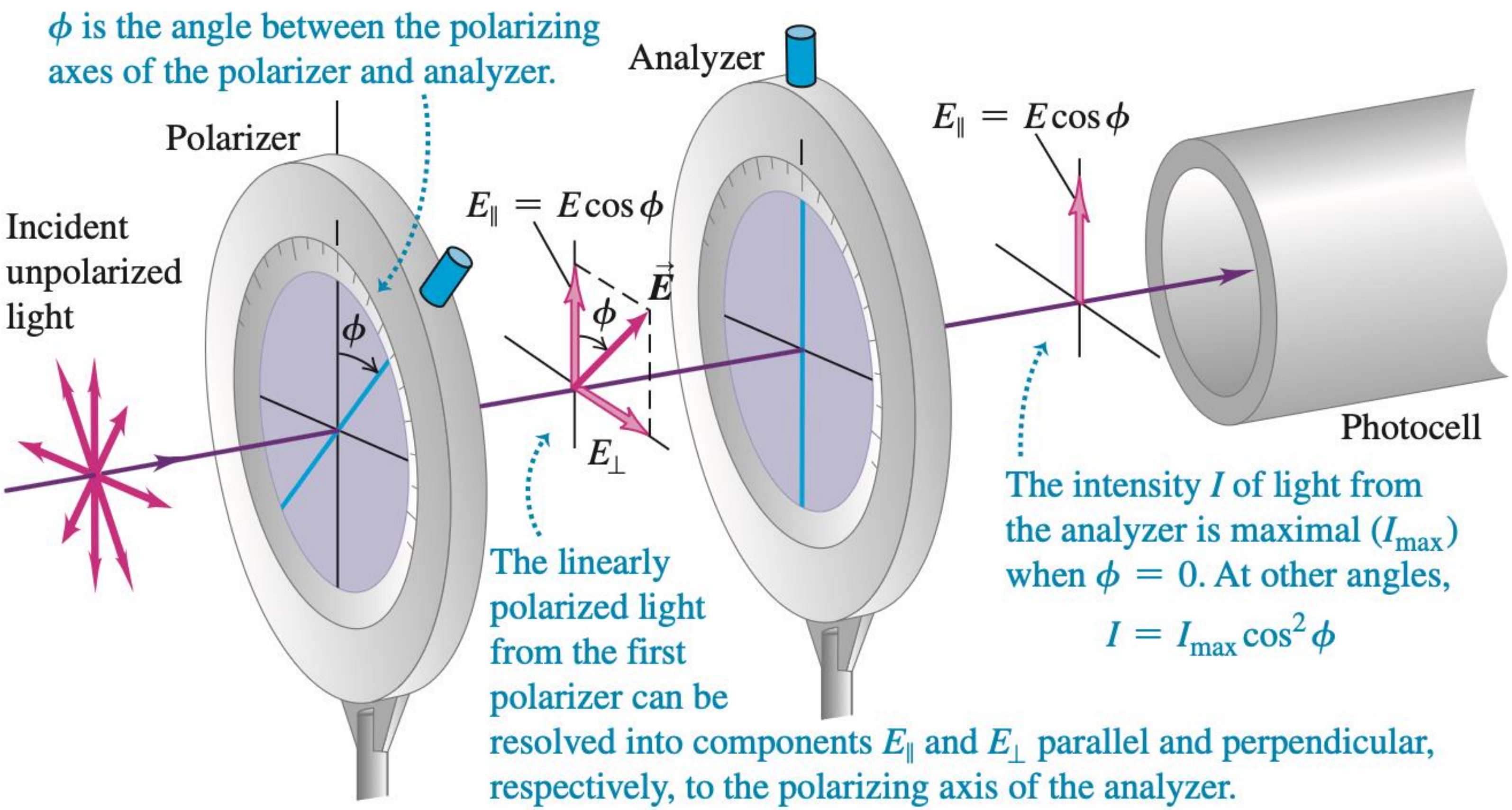
Please read 37.6 in Serway & Jewett

Polarization by Selective Absorption

- 1938, E.H. Land discovered *Polaroid*
- Material made of long strands of conductive hydrocarbons along which electrons can easily move.
- Energy from component of electric field parallel to molecules is absorbed.
- Transmission axis (polarizing axis) is perpendicular to the molecules.



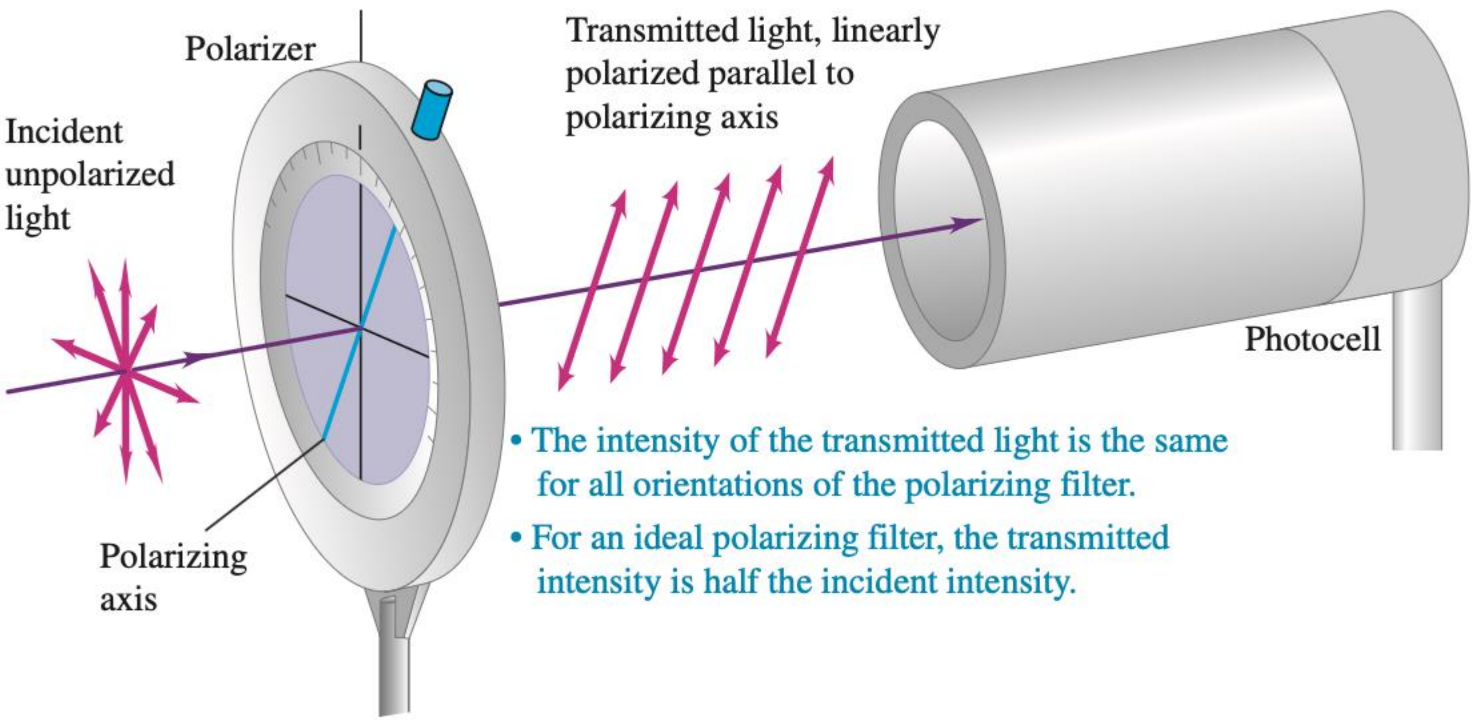
Malus's Law



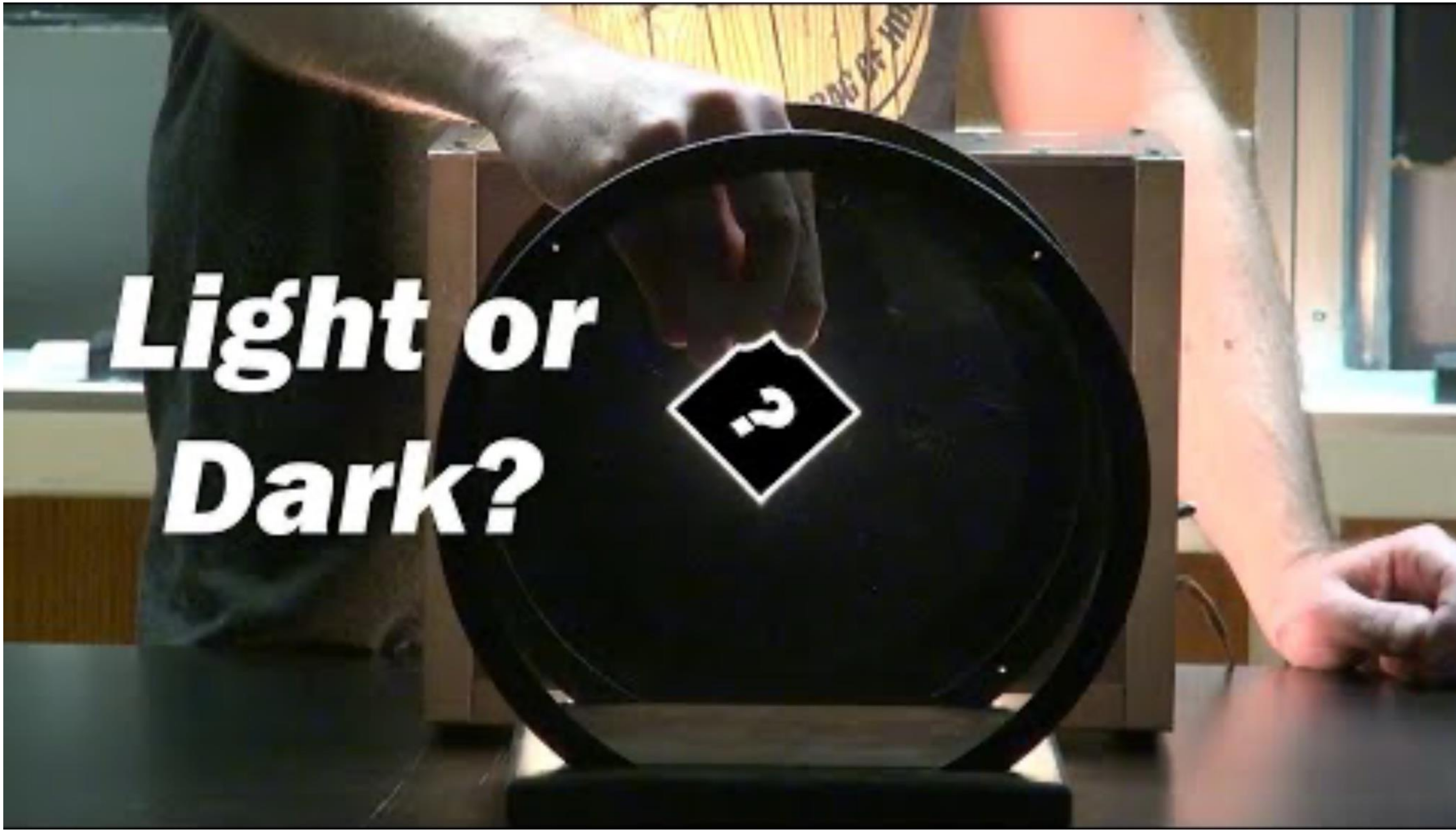
$$\boxed{I = I_{\max}\cos^2\phi}$$

Example - Malus's Law

Unpolarized light is incident on a polarizing filter. Compute the ratio of the intensity of transmitted light to the intensity of incident light.



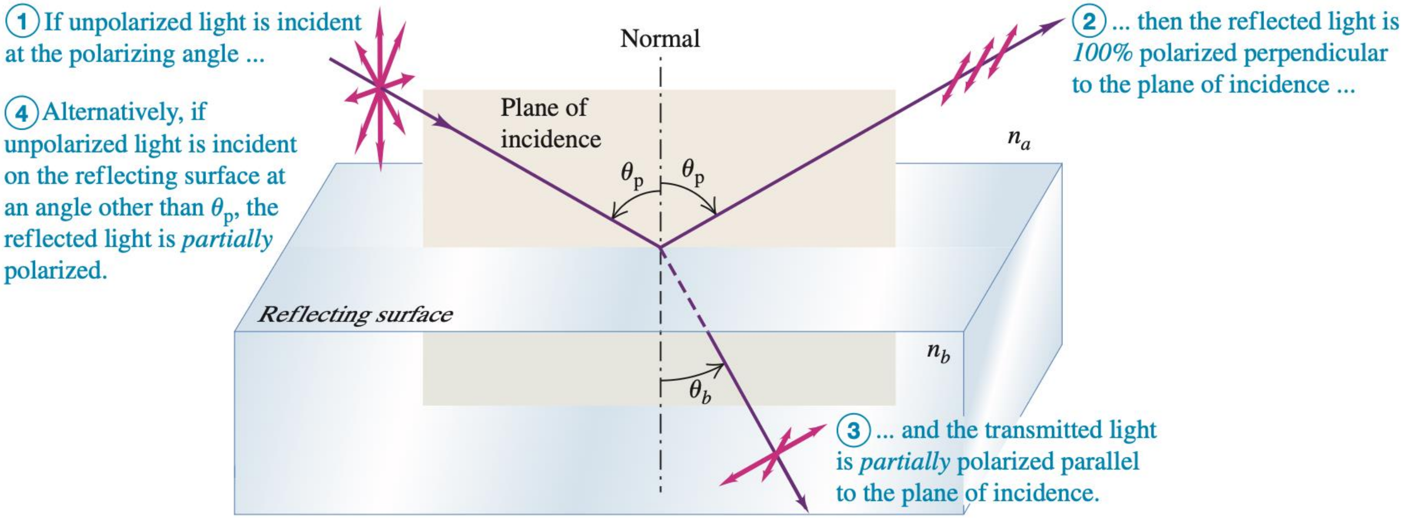
Demonstration

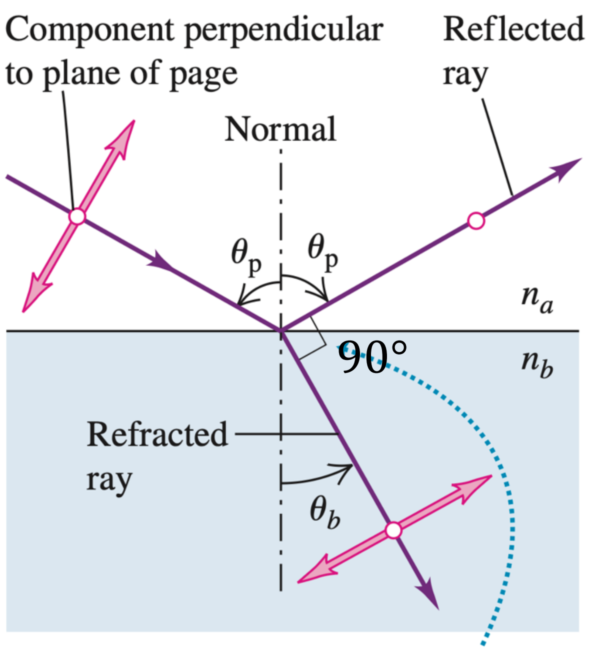


Polarization by Reflection

- Unpolarized light incident on a reflecting surface between two transparent optical materials can be partially or completely polarized in direction perpendicular to plane of incidence in a way that depends primarily on the angle of incidence.
- At the polarization angle, reflected light is completely polarized in the direction perpendicular to plane of incidence.
- Physical explanation: incident light activates dipole radiation in the second medium. Oscillating dipoles do not radiate in the direction of oscillation.
  - This is why you shouldn't necessarily point your radio antennae at the transmission tower; for best reception, your antennae should be parallel to the transmission antennae.

Polarization Angle (Brewster's Angle)





$$\boxed{\tan \theta_P = \frac{n_2}{n_1}}$$

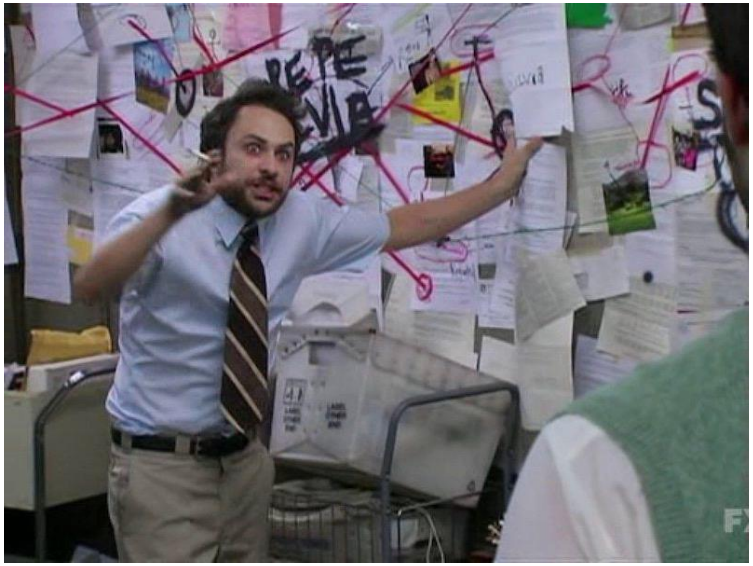
At the polarization angle, reflected light is completely polarized in the direction perpendicular to plane of incidence.

Demonstration

Rethinking Light and Matter

Modern quantum theory was a culmination of intense research that took place during the period from ~1900 to 1924.

- Classical physics was unable to explain the behavior of light and matter in the microscopic domain.
- Arbitrary rules were invented to make predictions match observations.
- We're not just talking about esoteric concerns of academics—*the very stability of matter was on the line*. How embarrassing!



^average physicist from early 20th century trying to explain why atoms don't fall apart in < 1 ps

Why Do Hot Objects Glow?

One of the earliest failures of classical physics was the inability to explain why hot objects glow. When heated, a solid object emits radiation across a continuous distribution of wavelengths (a spectrum). As the temperature increases, the object becomes red, then yellow, then white.

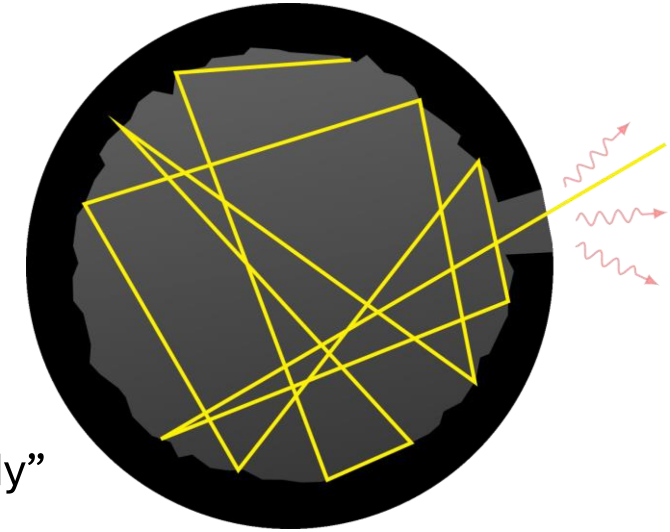
Ideal Blackbody

An ideal blackbody is an object that absorbs all incident light, reflecting none.

- An object in thermal equilibrium with its surroundings radiates as much energy as it absorbs.
- Hence, a blackbody is a perfect emitter as well as a perfect absorber of radiation.

The spectrum of a blackbody is universal, meaning it is independent of material composition.

- Provides insight into fundamental interactions between light and matter.



"cavity blackbody"

Blackbody Radiation

<u>Empirical Findings:</u>

1. Stefan-Boltzmann law: total power of emitted radiation given by

$$\boxed{P = \sigma A \epsilon T^4}$$

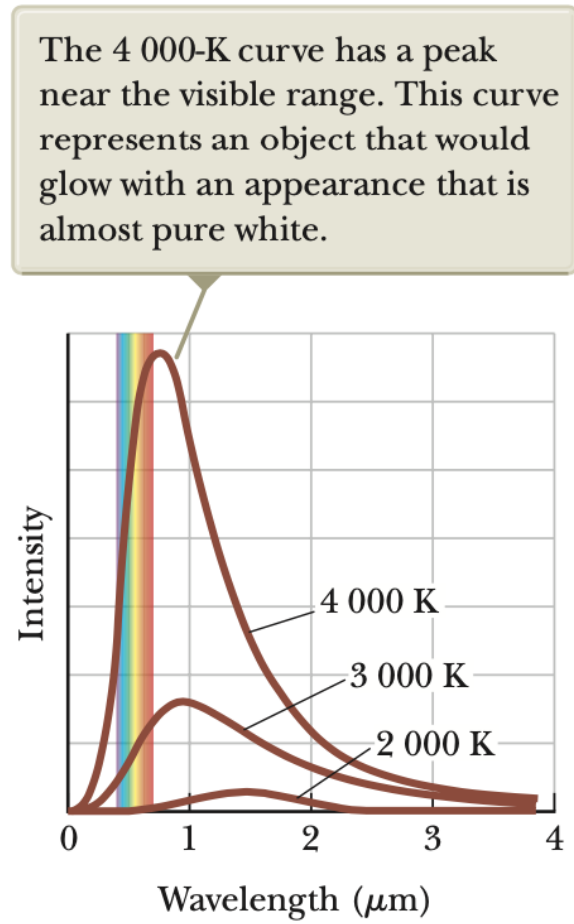
$$\sigma = 5.67 \times 10^{-8} \frac{\mathrm{W}}{\mathrm{m}^2 \cdot \mathrm{K}^4}$$

ϵ: emissivity between 0 and 1

A: surface area

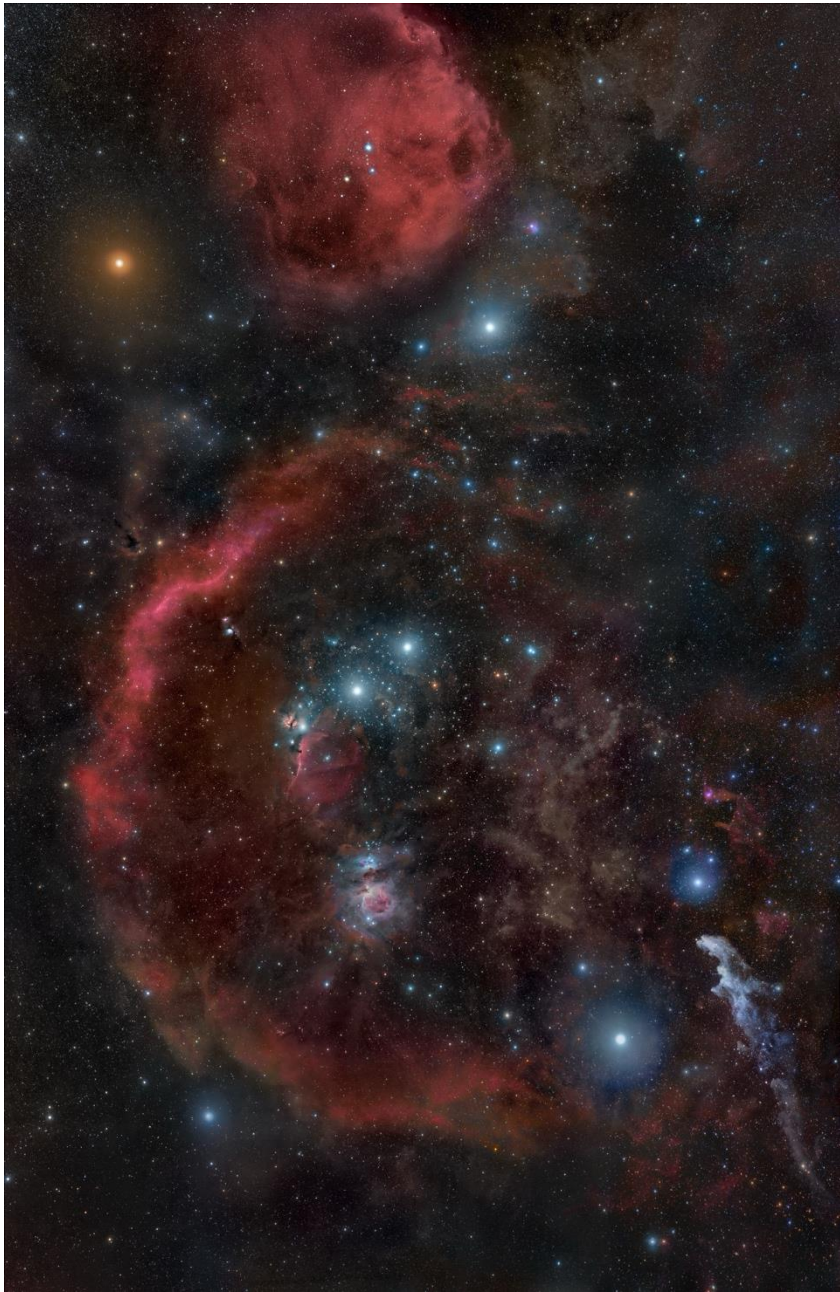
2. Wien's law: the peak of the spectrum shifts to shorter wavelengths as temperature increases.

$$\lambda_{\mathrm{max}} T = 2.898 \times 10^{-3}\ \mathrm{m} \cdot \mathrm{K}$$



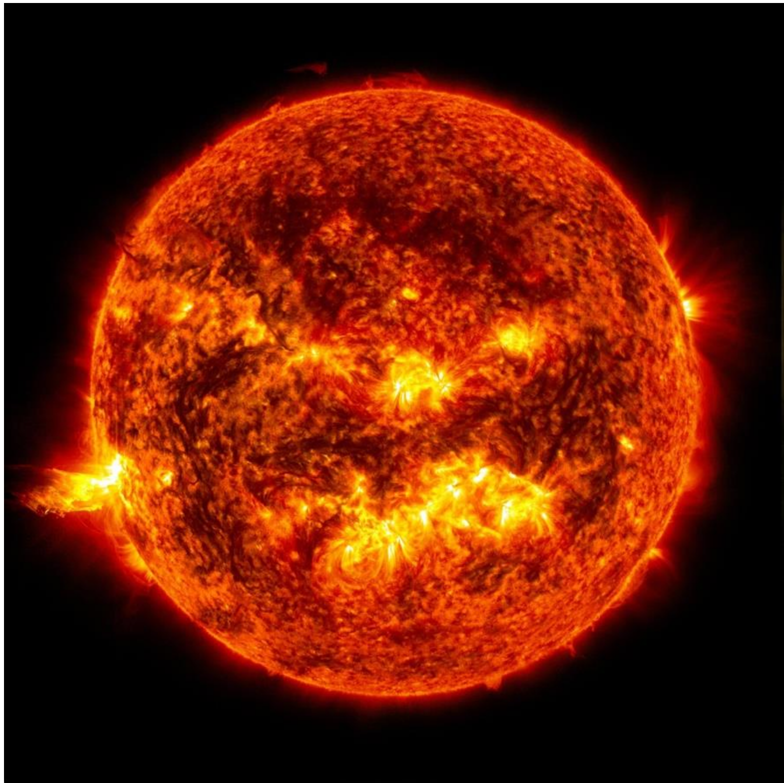
Example 1

Measurement of the wavelength at which the spectral distribution from a certain star is maximum indicates that the star's surface temperature is 3500 K. If the star is also found to radiate 100 times the power radiated by the Sun, how big is the star? The Sun's surface temperature is 5800 K.



Example

The surface temperature of the Sun is about 5800 K, and measurements of the Sun's spectral distribution show that it radiates very nearly like a blackbody, deviating mainly at very short wavelengths. Assuming the Sun radiates like an ideal blackbody, at what wavelength does the peak of the solar spectrum occur?



Blackbody Spectrum

To describe the distribution of energy from a black body, we define $u_f(f,T)df$ to be the energy density (energy per unit volume) of radiation in the frequency interval f to $f + df$.

The total energy density is

$$u = \int_0^\infty \underbrace{u_f(f,T)}_{\text{Energy density per unit frequency}}\, df$$

“spectral energy density”

Spectral Energy Density

Since $f = c/\lambda$, we can also write the spectral energy density in terms of the wavelength:

$$u = \int_0^\infty u_f(f,T)\,df = \int_0^\infty u_f(f(\lambda),T)\cdot\frac{c}{\lambda^2}\,d\lambda$$

$$u_\lambda(\lambda,T) = u_f(f(\lambda),T)\cdot\frac{c}{\lambda^2}$$

$$u = \int_0^\infty u_\lambda(\lambda,T)\,d\lambda$$

Spectral Intensity

Spectral intensity (or specific intensity) is defined as the intensity per unit wavelength or frequency. The relationship between spectral intensity and spectral energy density depends on the nature of the radiation.

$$I = \int_0^{\infty} I_\lambda(\lambda, T)\, d\lambda$$

For isotropic radiation, spectral intensity I_λ is directly proportional to the spectral energy density u_λ:

$$I_\lambda = \frac{1}{4} c\, u_\lambda$$

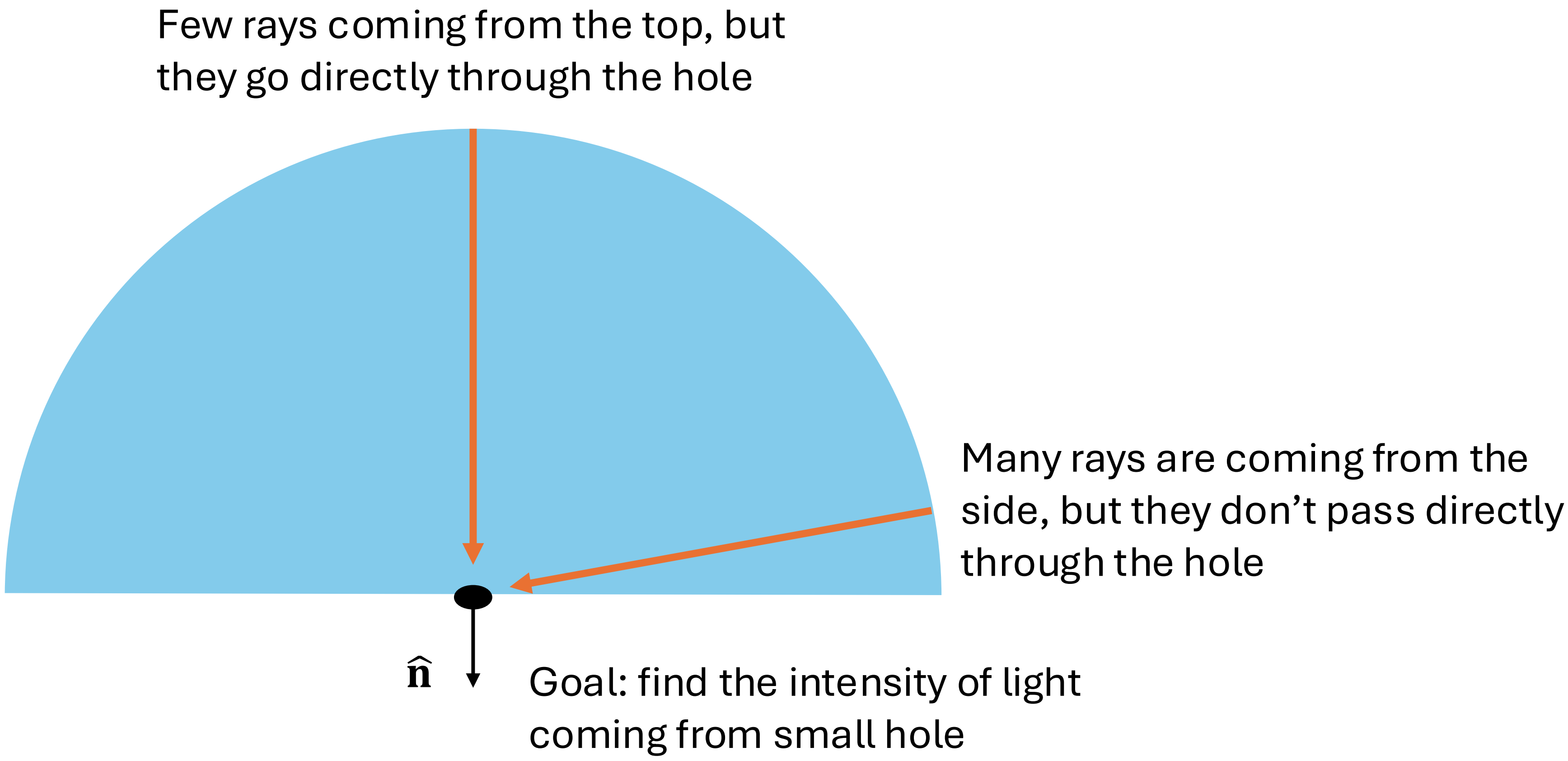
Sticking to “per unit wavelength” since that’s what your textbook uses

Why the factor of $c/4$?

If all radiation is moving in one direction, intensity is simply $I = cu$ (recall the Poynting vector from PHYS 152).

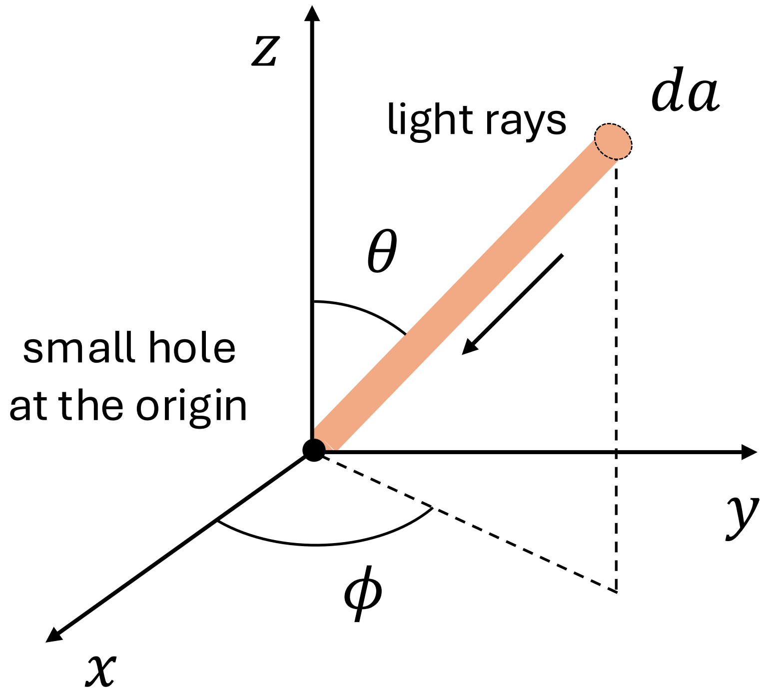
The intensity at each point in a region with isotropic radiation is reduced by ¼ for two reasons:

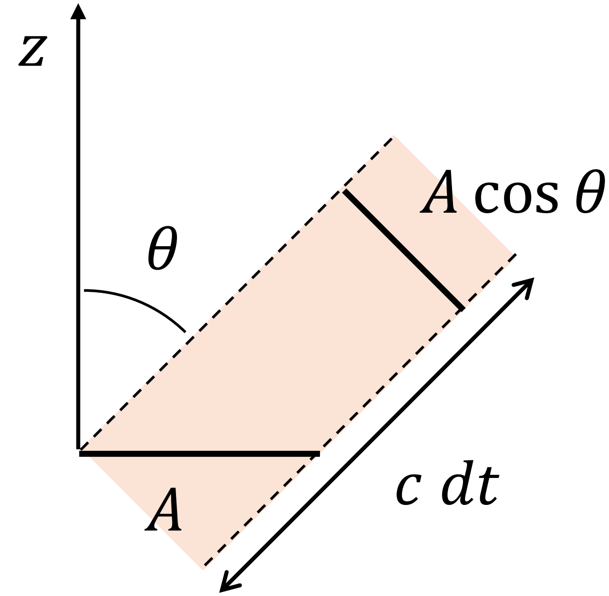
1. Only $1/2$ the radiation is outward with nonzero component parallel to some direction $\hat{\mathbf{n}}$.
2. Most outward rays are oblique. A ray at angle θ to $\hat{\mathbf{n}}$ contributes a factor $\cos\theta$ to the flux because of projection. When you average $\cos\theta$ over a hemisphere, you get another factor of $1/2$.



$$\text{Intensity} \propto \#\text{ of rays } \times \text{ flux parallel to } \hat{\mathbf{n}}$$

Why the factor $c/4$?





$$dE_\lambda = \frac{\text{energy density}}{\text{wavelength}} \times \text{ volume}$$

$$dE_\lambda = \left(u_\lambda \cdot \frac{da}{4\pi r^2}\right) \times \ (A\cos\theta\ \ c dt)$$

$$E_\lambda = \int_{\text{upper hemisphere}} dE_\lambda$$ "spectral energy"

$$E_\lambda = \int u_\lambda \frac{da}{4\pi r^2} \; A\cos\theta \; c dt$$

$$E_\lambda = u_\lambda \frac{c\ A\ dt}{4\pi r^2} \int_0^{2\pi} \int_0^{\frac{\pi}{2}} \cos\theta \; r^2 \sin\theta \; d\theta\ d\phi$$

$$E_\lambda = u_\lambda \frac{c\ A\ dt}{4}$$

$$I_\lambda = \frac{E_\lambda}{A\ dt} = \frac{c}{4} u_\lambda$$

Rayleigh-Jeans Law

Classical E&M and statistical mechanics can be applied to a cavity containing standing EM waves to **derive** the Rayleigh-Jeans law

$$u_f(f,T) = \frac{8\pi f^2}{c^3}\, k_B T$$

- $k_B = 1.38 \times 10^{-23}\ \mathrm{J/K} = 8.617 \times 10^{-5}\ \mathrm{eV/K}$
- The Rayleigh-Jeans law works well for low frequencies, but it diverges at high frequencies
- Furthermore, it predicts infinite energy density, and hence infinite total energy!

"Ultraviolet catastrophe"

Rayleigh-Jeans

$$u_f(f,T) = \frac{8\pi f^2}{c^3}\,k_B T$$

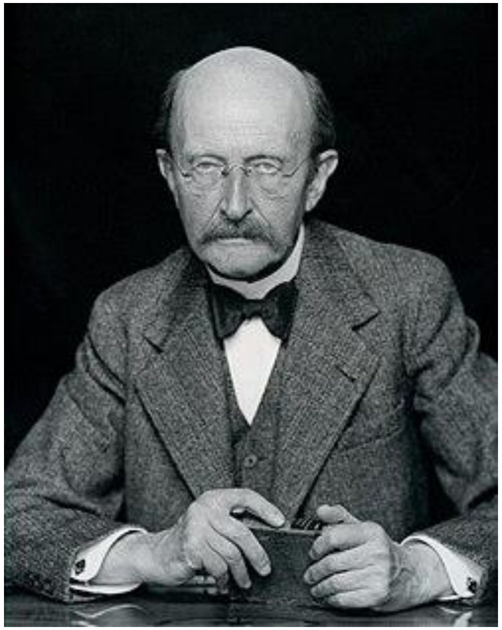
$$u_\lambda(\lambda,T) = \frac{8\pi}{\lambda^4}\,k_B T$$

$$I_\lambda(\lambda,T) = \frac{2\pi c}{\lambda^4}\,k_B T$$

Eq. 39.3 in
Serway & Jewett

Planck's Math Trick

- Rayleigh-Jeans assumes energy exchange between radiation and matter is *continuous*.
- In 1900, Max Planck derived an accurate equation for blackbody spectrum by employing a "math trick"
  - Calculated the average energy by considering small but discrete energy exchange between radiation and the walls of the cavity.
  - Purely formal assumption, didn't believe in physical reality of discrete energy exchange at first.



Planck's Distribution

Planck found the energy density per unit frequency of the radiation emitted from a blackbody is

$$u_f(f,T) = \frac{8\pi f^2}{c^3}\frac{hf}{e^{hf/k_BT}-1}$$

- Fits the data *exactly* as long as

$$h = 6.626 \times 10^{-34}\,\mathrm{J\cdot s}$$
$$= 4.136 \times 10^{-15}\,\mathrm{eV\cdot s}$$

- Predicts finite total energy, matches Rayleigh-Jeans in the low frequency limit, and leads to the empirical findings of Stefan and Wien.

See supplementary document on Brightspace titled "Blackbody radiation"

Planck's Distribution

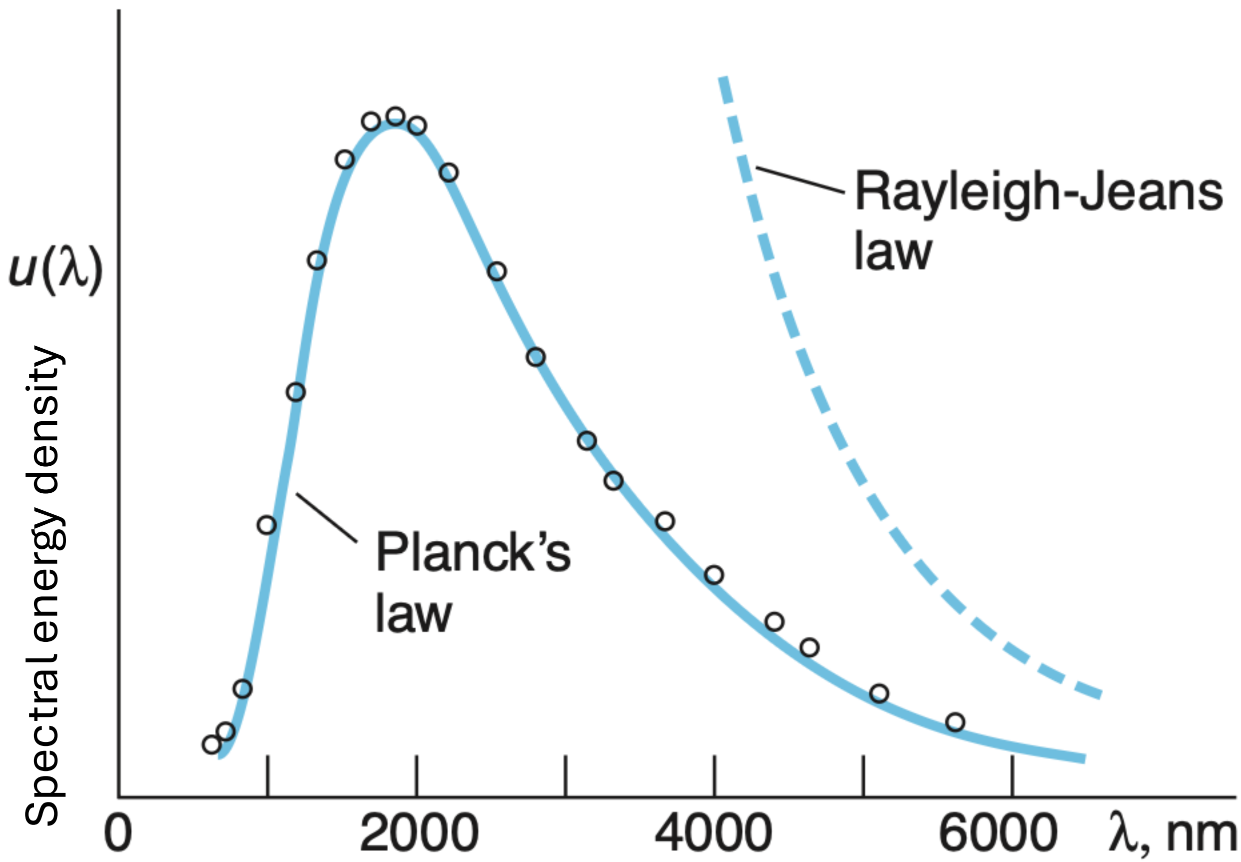
$$u_f(f,T) = \frac{8\pi}{\lambda^3} \frac{h}{e^{hf/k_BT} - 1}$$

$$u_\lambda(\lambda,T) = \frac{8\pi}{\lambda^5} \frac{hc}{e^{hf/k_BT} - 1}$$

$$I_\lambda(\lambda,T) = \frac{2\pi c}{\lambda^5} \frac{hc}{e^{hf/k_BT} - 1}$$

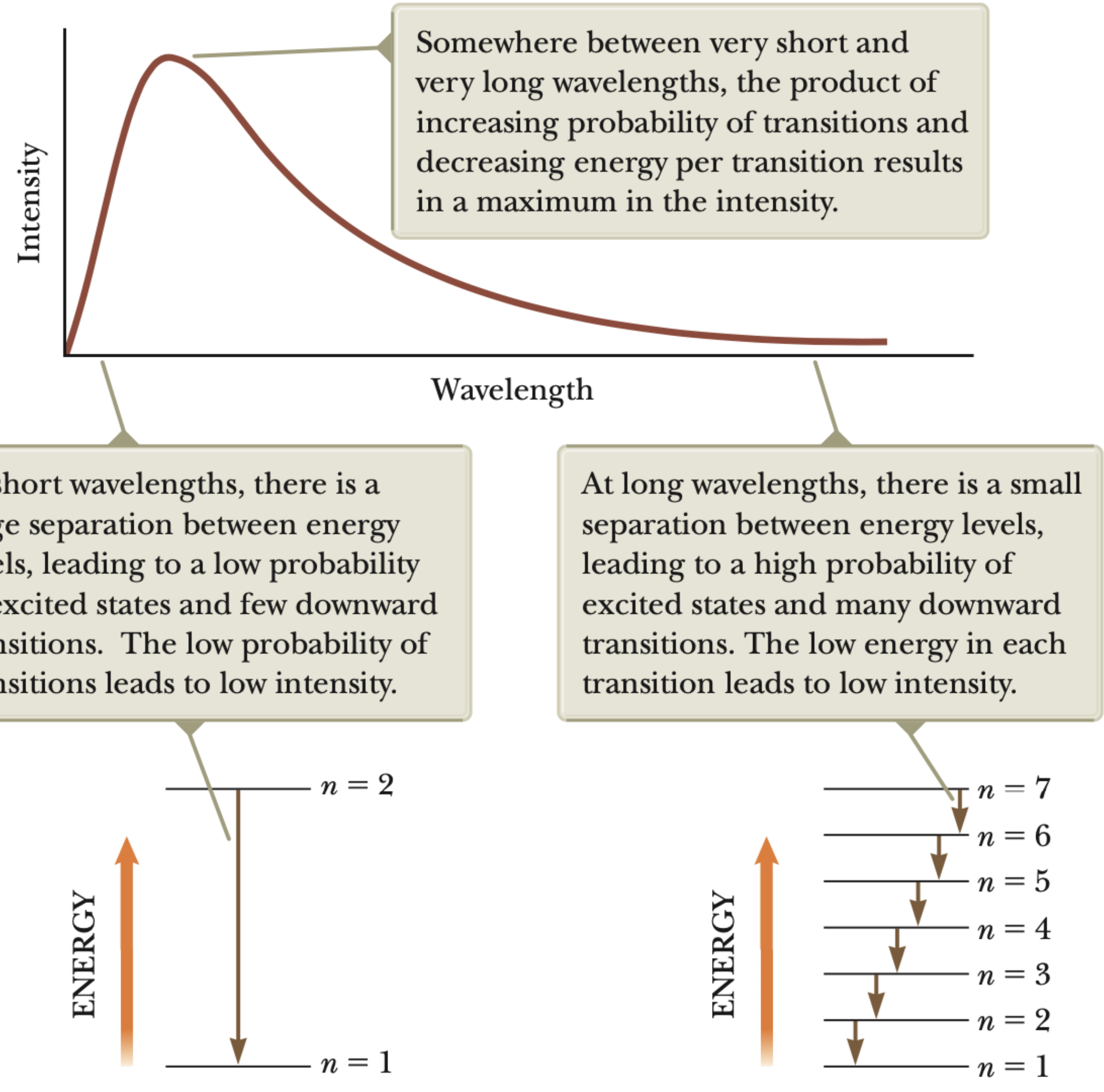
Eq. 39.6 in
Serway & Jewett

Blackbody Spectrum



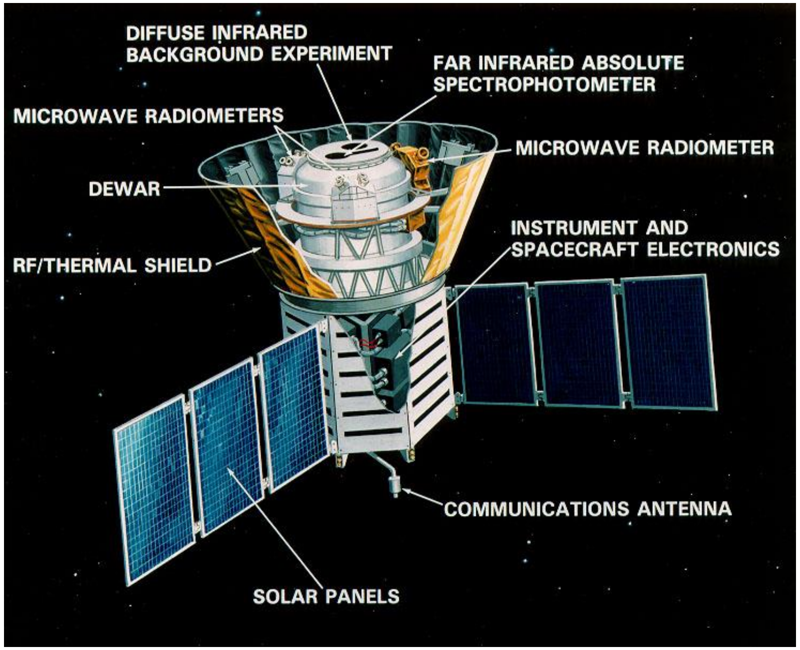
Comparison of Planck's law and the Rayleigh-Jeans equation with experimental data at $T = 1600$ K obtained by Coblenz in 1915.

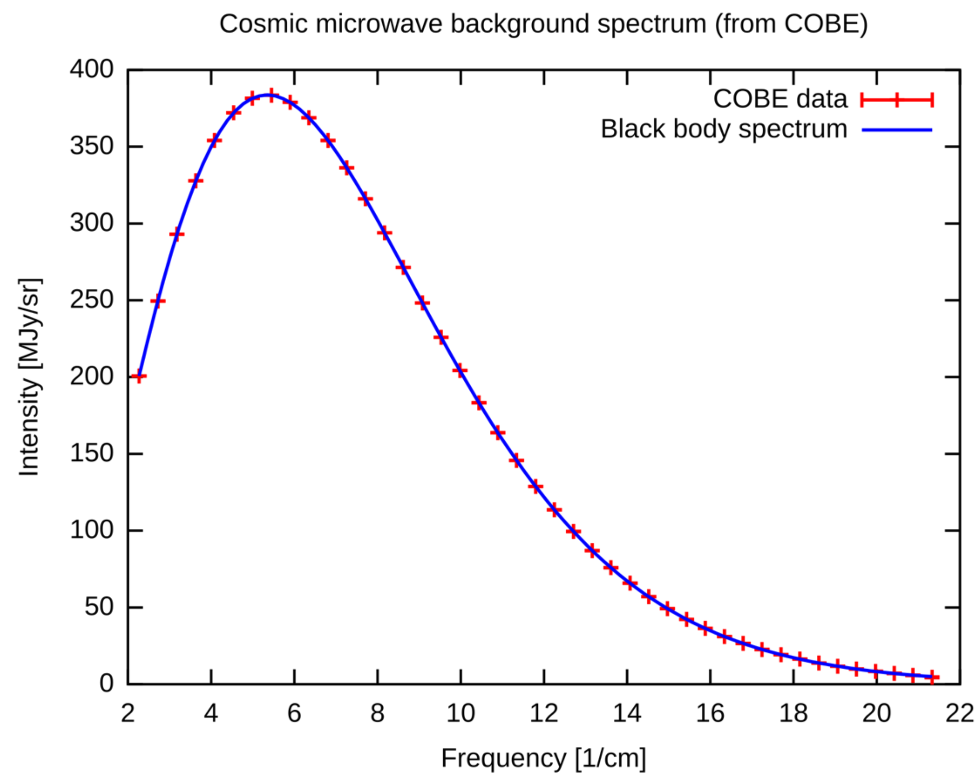
Planck's Distribution



Application of Planck’s Law

- Early universe was hot, dense, and in thermal equilibrium. Radiation followed **Planck’s blackbody spectrum**.
- As the universe expanded, radiation cooled but **kept the Planck form** (temperature just decreased).
- Big Bang prediction: a **uniform, isotropic background radiation** with a blackbody spectrum at just a few kelvin.
- **Penzias & Wilson (1965)** detected this radiation as an unexplained ~3 K microwave signal present in all directions.
- Later measurements (especially **COBE**) showed the **spectral energy density matches Planck’s law extremely precisely** (~2.725 K blackbody).
- This is strong evidence that the universe was once in a **hot, thermal equilibrium state**. Also considered strong support for the Big Bang theory itself!


DIFFUSE INFRARED
BACKGROUND EXPERIMENT
FAR INFRARED ABSOLUTE
SPECTROPHOTOMETER
MICROWAVE RADIOMETERS
MICROWAVE RADIOMETER
DEWAR
INSTRUMENT AND
SPACECRAFT ELECTRONICS
RF/THERMAL SHIELD
COMMUNICATIONS ANTENNA
SOLAR PANELS


Cosmic microwave background spectrum (from COBE)
COBE data
Black body spectrum
Intensity [MJy/sr]
Frequency [1/cm]
0
50
100
150
200
250
300
350
400
2
4
6
8
10
12
14
16
18
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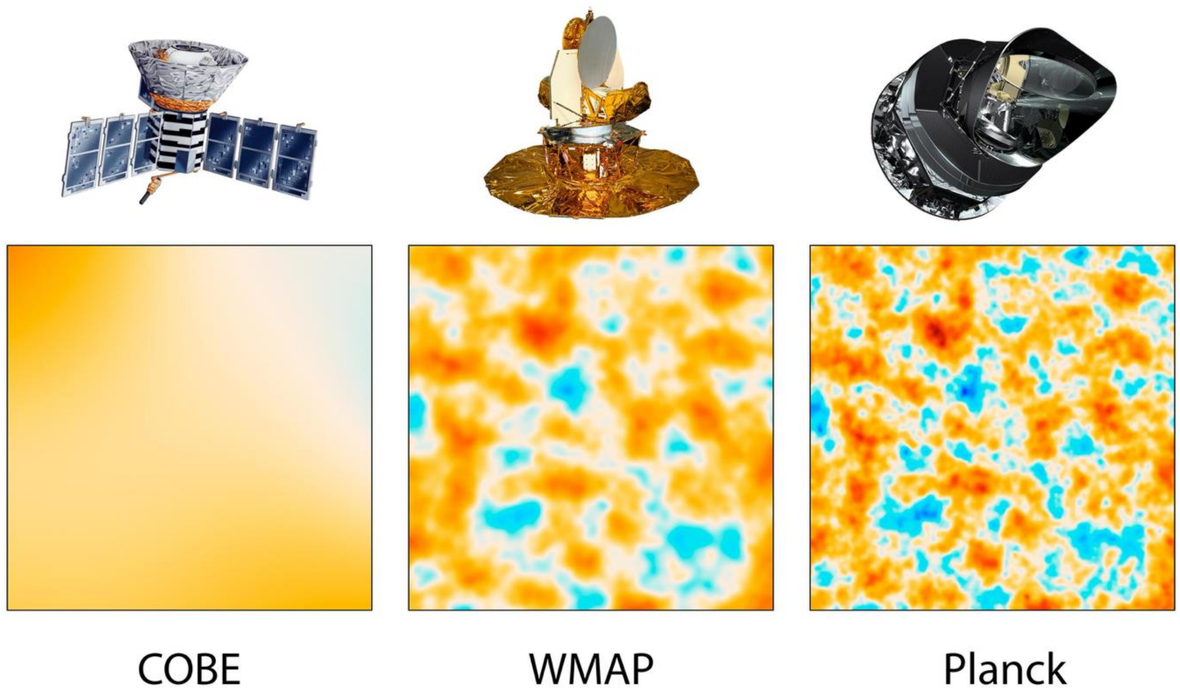


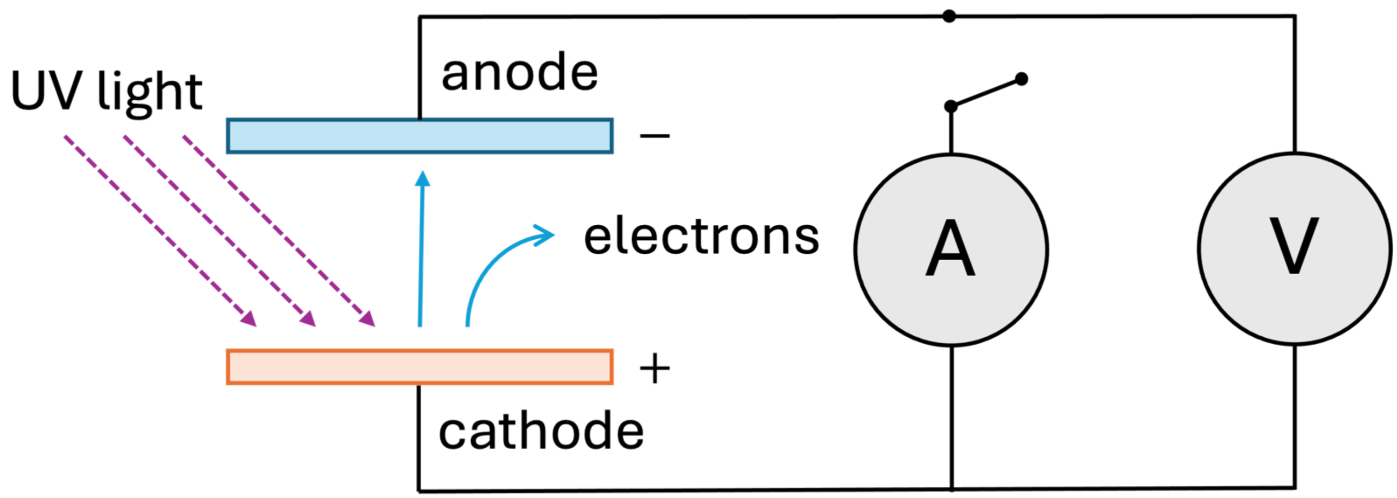
Anisotropies (tiny temperature fluctuations, $\Delta T/T \sim 10^{-5}$) important because:

- **Initial seeds of structure** (galaxies, clusters)
- They encode detailed physics of the early universe
- Density of matter/dark matter, geometry, expansion

Demonstration

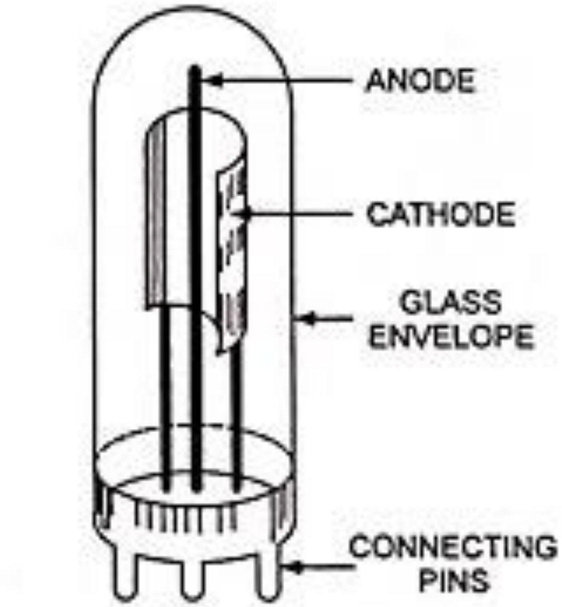
Photoelectric Effect

First discovered by Heinrich Hertz in 1887, the photoelectric effect describes the ability of light to dislodge electrons from a metallic surface.



$$K_{\text{max}} = e\,\Delta V_S$$

Phototube



Photoelectric Effect

List of experimental findings:

1. At high frequencies and intensities, rate of ejected electrons is proportional to intensity of light.
2. Even at low intensities, electrons are ejected *essentially instantly* with delays less than 10 ns.
3. If the intensity is held constant, the rate of ejected electrons decreases with increasing frequency in a similar way for all metals.

Photoelectric Effect

Experimental findings continued:

4. If the frequency is below a material-dependent cutoff frequency f_c , no electrons are ejected, no matter how high the intensity is.
5. If the frequency is held constant, the maximum kinetic energy of electrons $K_{\max}$ is independent of the light's intensity.
6. Above the cutoff frequency, $K_{\max}$ is directly proportional to the frequency with the same slope for all metals.

Wave Model Falls Flat

According to the classical wave model of radiation,

1. At high intensities, the rate at which electrons are ejected will be proportional to the intensity of the light.
2. At low intensities, there will be a significant delay between the illumination of the plate and the emission of electrons.
3. The rate of electron ejection depends in a metal-specific way on the frequency of the light (due to resonance).
4. The maximum kinetic energy of the ejected electrons will increase with increasing intensity (and may also depend on the light's frequency due to resonance).

In summary, the wave model is unable to account for many of the experimentally verified features of the photoelectric effect!

Photon Model of Light

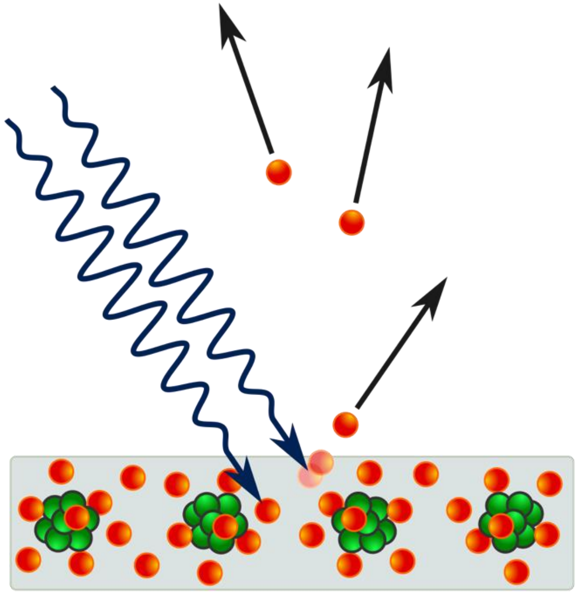
The photon model of light and its application to the photoelectric effect was first proposed by Albert Einstein in 1905 and later confirmed by Roger Millikan in 1917.

- Each photon of incident light carries energy hf where f is the frequency of the associated light wave.
- Each electron is ejected from the metal as a result of collision (and subsequent absorption of) a single photon of light.

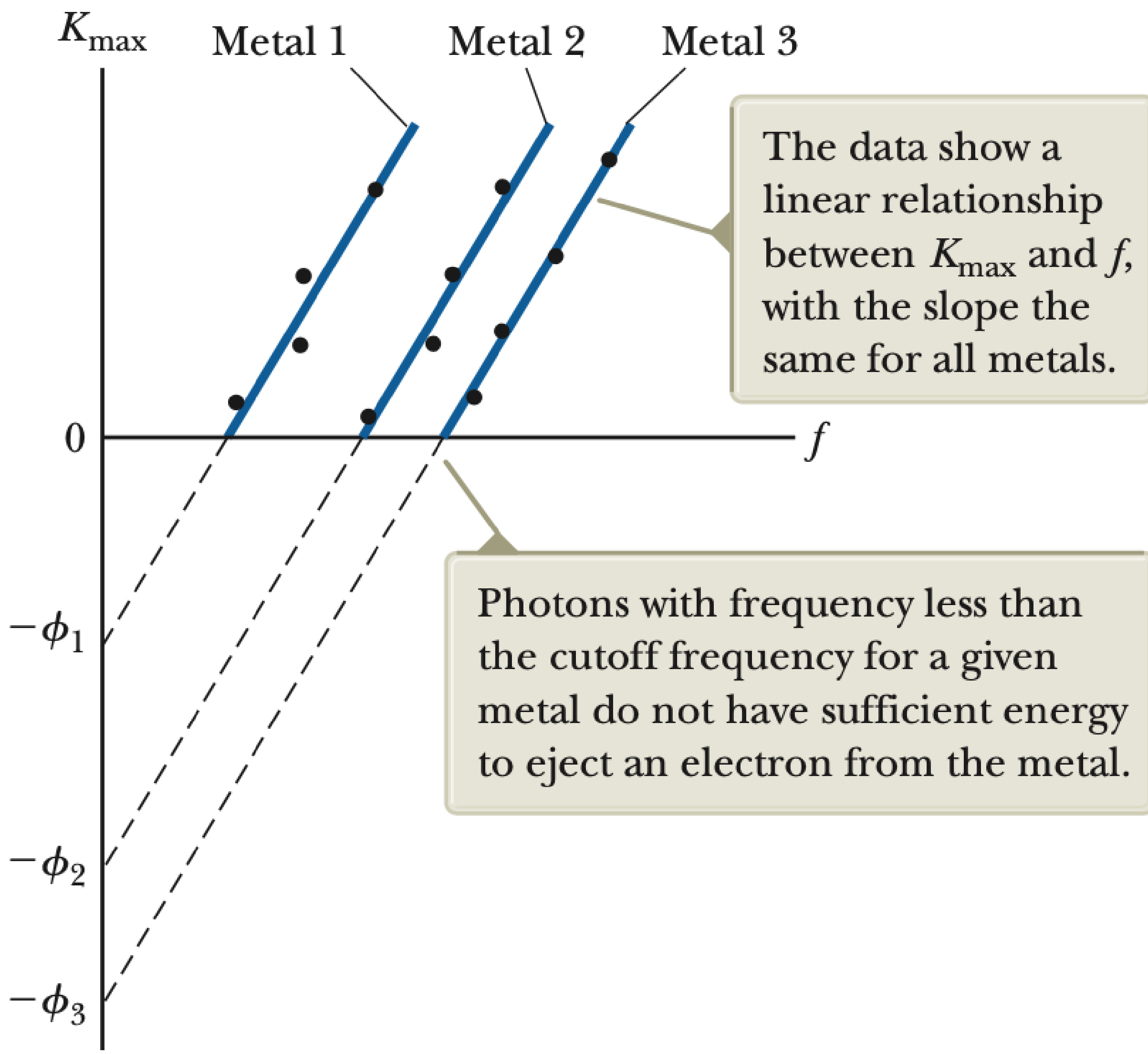
Maximum kinetic energy of electrons:

$$K_{\max} = hf - \phi$$

- ϕ is the **work function** of the metal – the minimum energy required to free an electron (on the order of electron-volts).



Photoelectric Effect



Example

A sodium surface is illuminated with light having a wavelength of 300 nm. As indicated in Table 39.1, the work function for sodium metal is 2.46 eV.

(a) Find the maximum kinetic energy of the ejected photoelectrons.

(b) Find the cutoff wavelength for sodium.

Example 2: Photoelectric Effect

When ultraviolet light of wavelength 250 nm falls on an aluminum surface, the measured maximum kinetic energy of ejected electrons is 0.88 eV, where 1 eV (electron-volt) is the energy required to accelerate a single electron from rest through a potential difference of 1 volt. When light of wavelength 210 nm falls on the same plate, the maximum kinetic energy is 1.82 eV. Check that the value of hc implied by these data is 1240 eV $\cdot$ nm and find the value of the work function for aluminum.

Recall that for electromagnetic radiation, $c = \lambda f$.

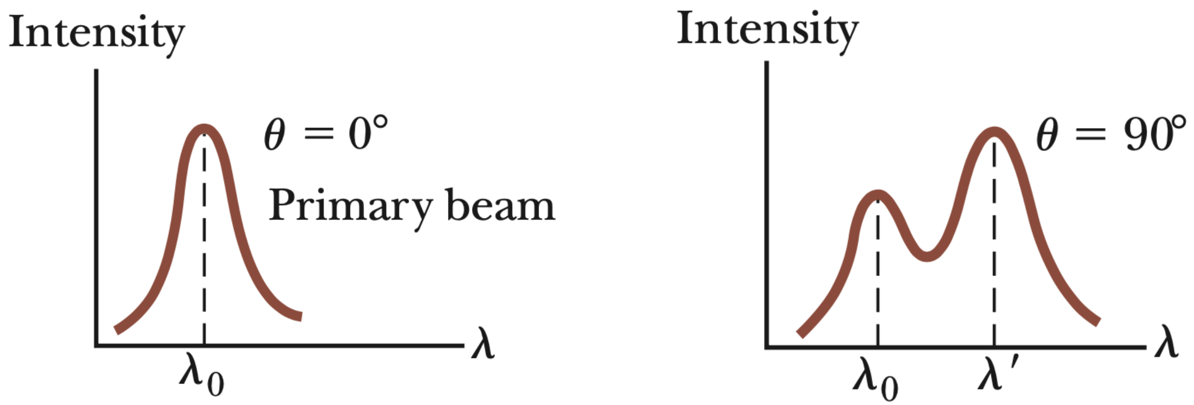
Example 3: Photon Model

A 100-W incandescent light bulb uses 100 W of electrical power but only radiates about 10 W of actual visible light. Estimate the number of photons that strike your face per second when you stand 2 m away from the light bulb.

Compton Effect

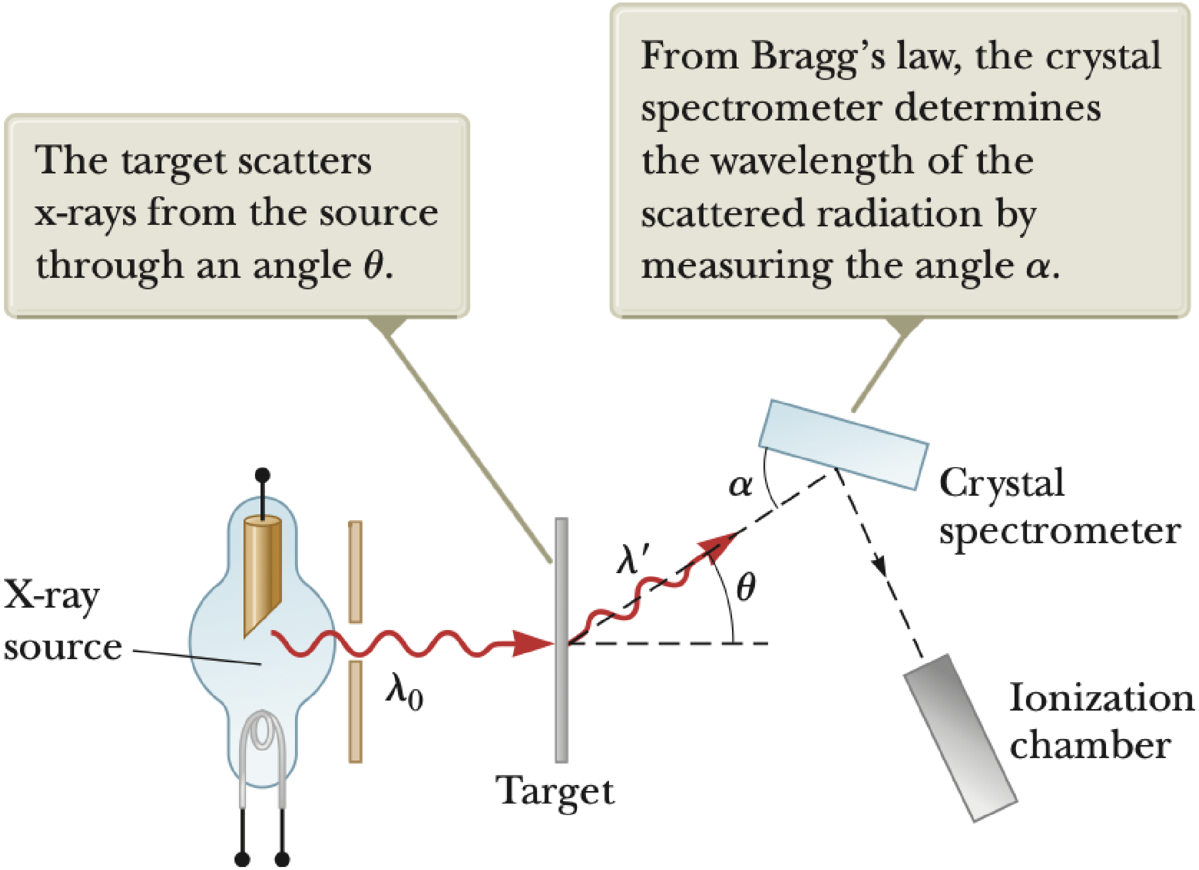
In 1923, Compton provided support for the particle aspect of radiation.

- Scattered X-rays off free electrons (frequency too high for absorption)
- Found that wavelength of scattered radiation is larger than wavelength of incident radiation, in a way that depends on the angle of the scattered radiation

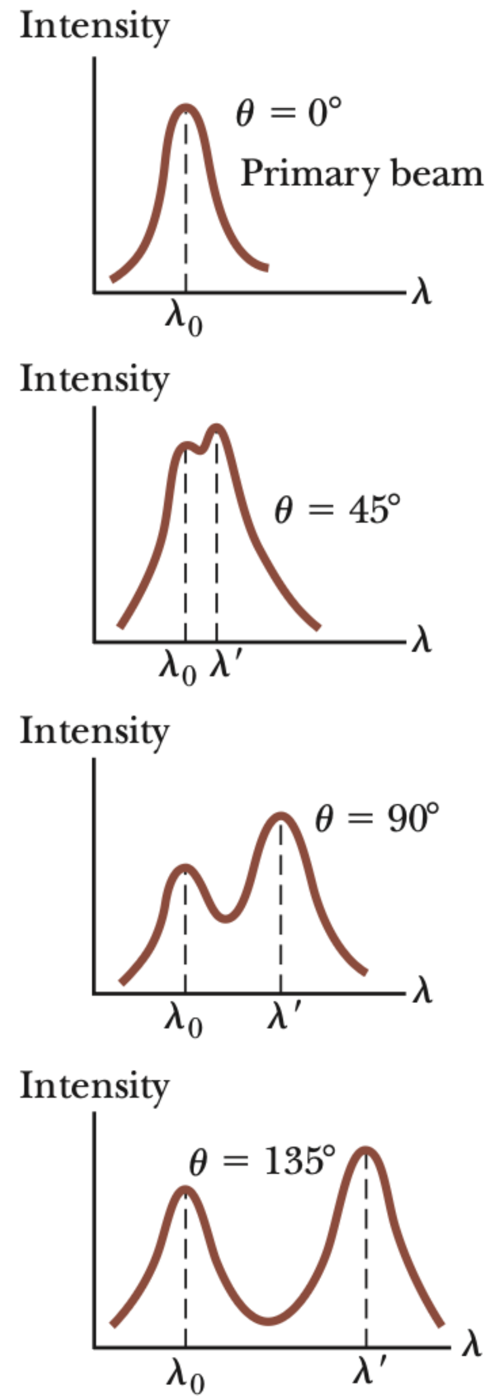


Wavelength shift is proportional to scattering angle.

Compton Effect



$$\lambda - \lambda' = \frac{hc}{m_e c^2}(1 - \cos\theta)$$



Wave-Particle Duality of Light

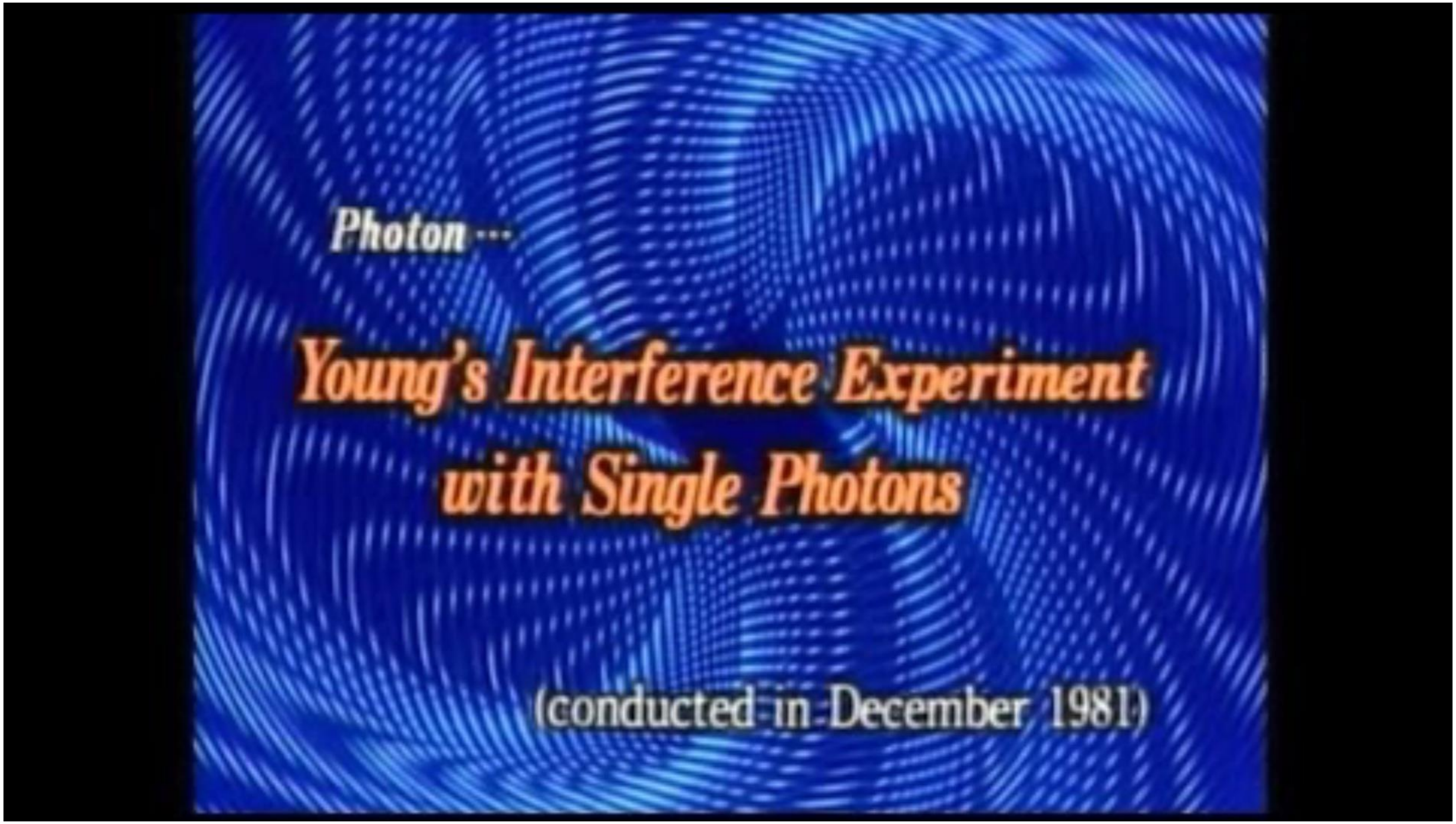
Within the framework of classical physics,

- Wave model: Young's double slit experiment, diffraction
- Photon model: photoelectric effect, Compton scattering

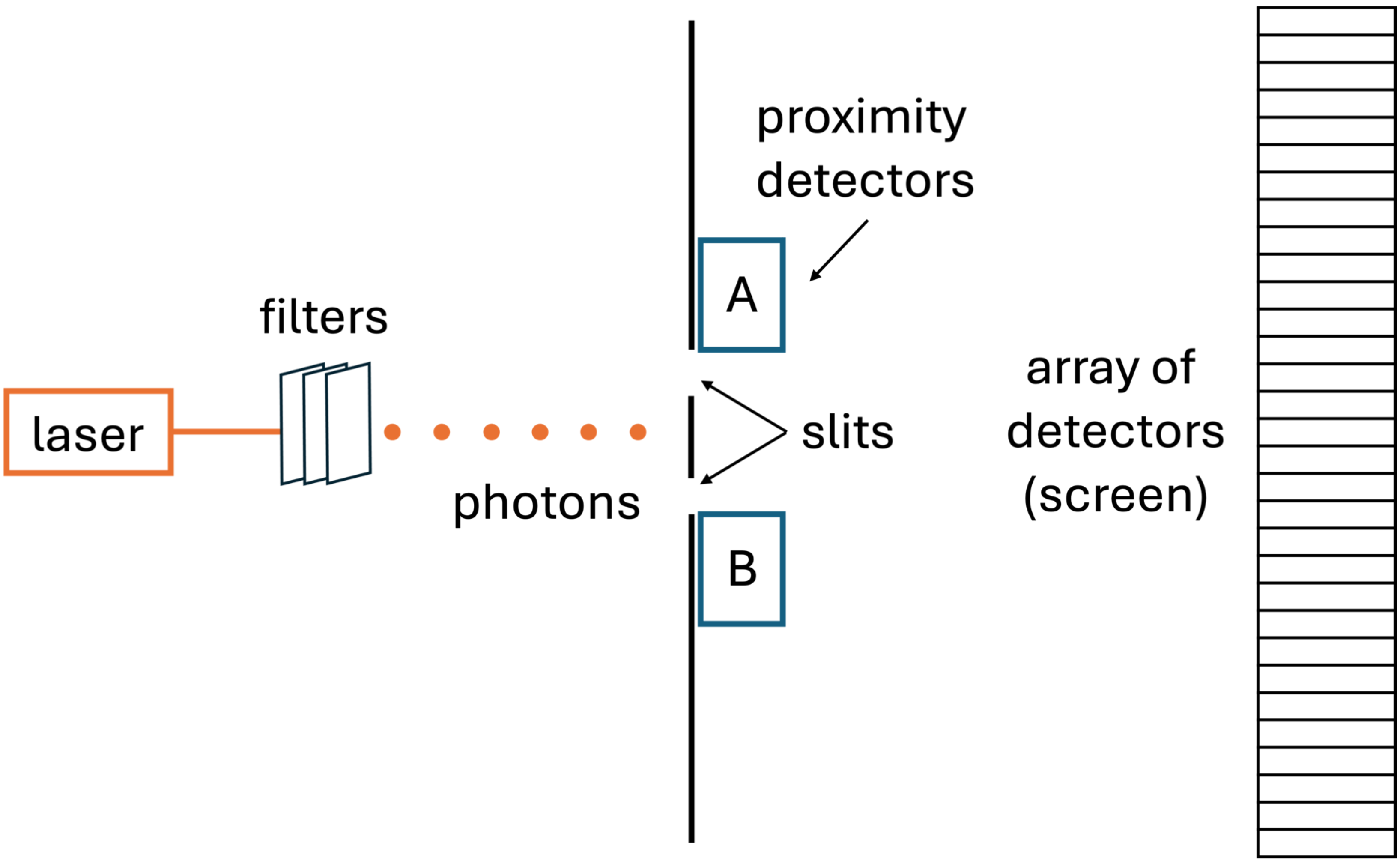
Maybe classical physics is alright; light just behaves like a wave sometimes and other times like a particle…

- Even in situations where we are almost certainly dealing with individual photons, light may still exhibit wave-like behavior, namely interference!
- In 1909, G.I. Taylor used a low intensity light source to demonstrate that even when individual photons pass through one by one through a double-slit, an interference pattern builds up over time.

Single-Photon Interference



Single-Photon Interference



Example 4: Single-Photon Source

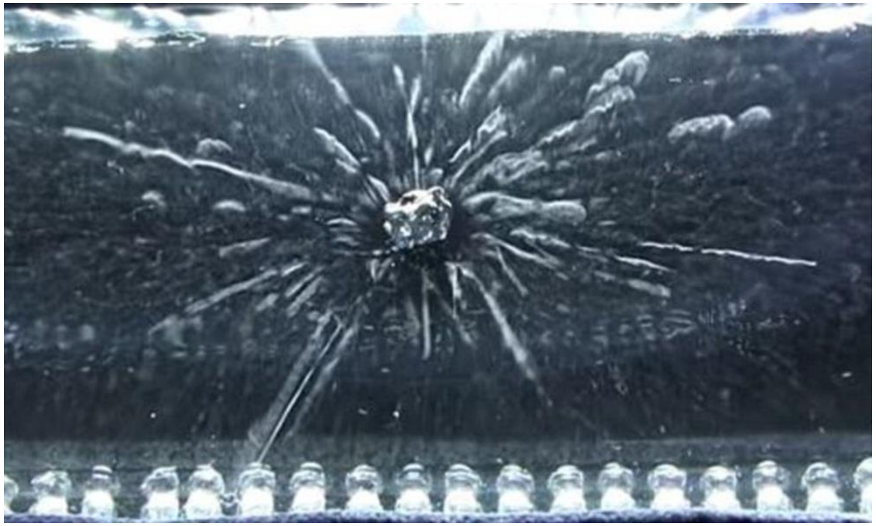
Assume that a laser emits about 10^{15} photons per second. If the distance between the filters and the detector array is about 3 m, then it will take a photon about 10^{-8} s to cover this distance. If we arrange things so that an average of one photon every 10^{-5} s (i.e., an average of 10^{5} photons per second) gets through the last filter, then there will be only a 1/1000 chance that a photon is in flight at any given time. So, we want to reduce the 10^{15} photons per second emitted by the laser to roughly 10^{5} photons per second emerging from the last filter.

If each filter transmits only 5% of incident light, how many filters are needed to create a single-photon source?

The Not-So-Atomic Atom

Atomos, the Greek word for atom, means indivisible.

- In the 1890's, radioactive materials were discovered, suggesting atoms have internal structure (they can fall apart).
- J.J. Thompson discovered β particles using cathode rays.
- Ernest Rutherford discovered α particles in the decay of uranium and other radioactive sources (like radon).

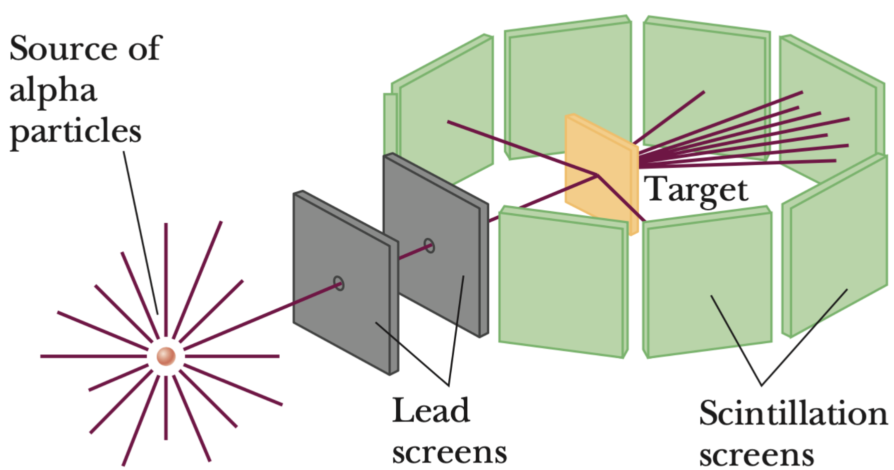


Uranium sample emitting particles into a cloud chamber.

Rutherford Scattering

In 1909, Rutherford began using α particles to probe atomic structure by directing them toward a gold foil.

- Most particles passed through with little-to-no deflection.
- On rare occasions (1 out of every 10^5 events) particles would scatter at a large angle, “as if you fired a 15-inch artillery shell at a piece of tissue paper and it came back and hit you.”



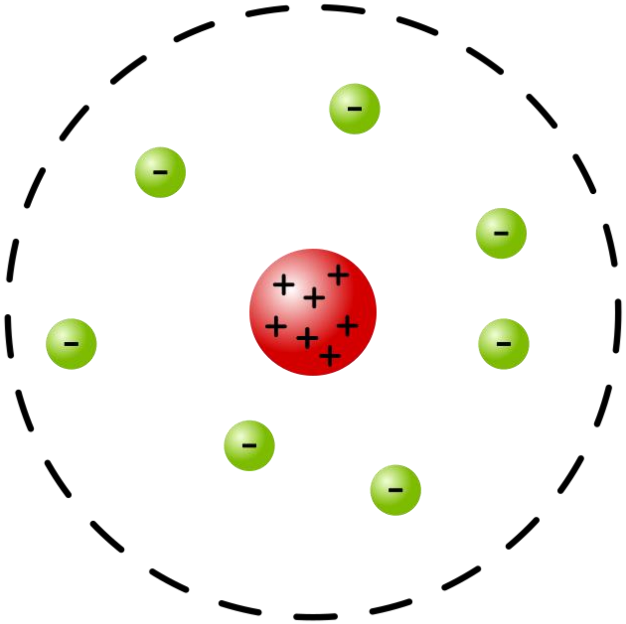
Rutherford's Planetary Model

Rutherford could only make sense of the scattering in the context of a "planetary model" where electrons orbit a very dense, positively charged nucleus.

Application of classical physics to this model predicts

i. Atoms should be unstable (with very short lifetimes).

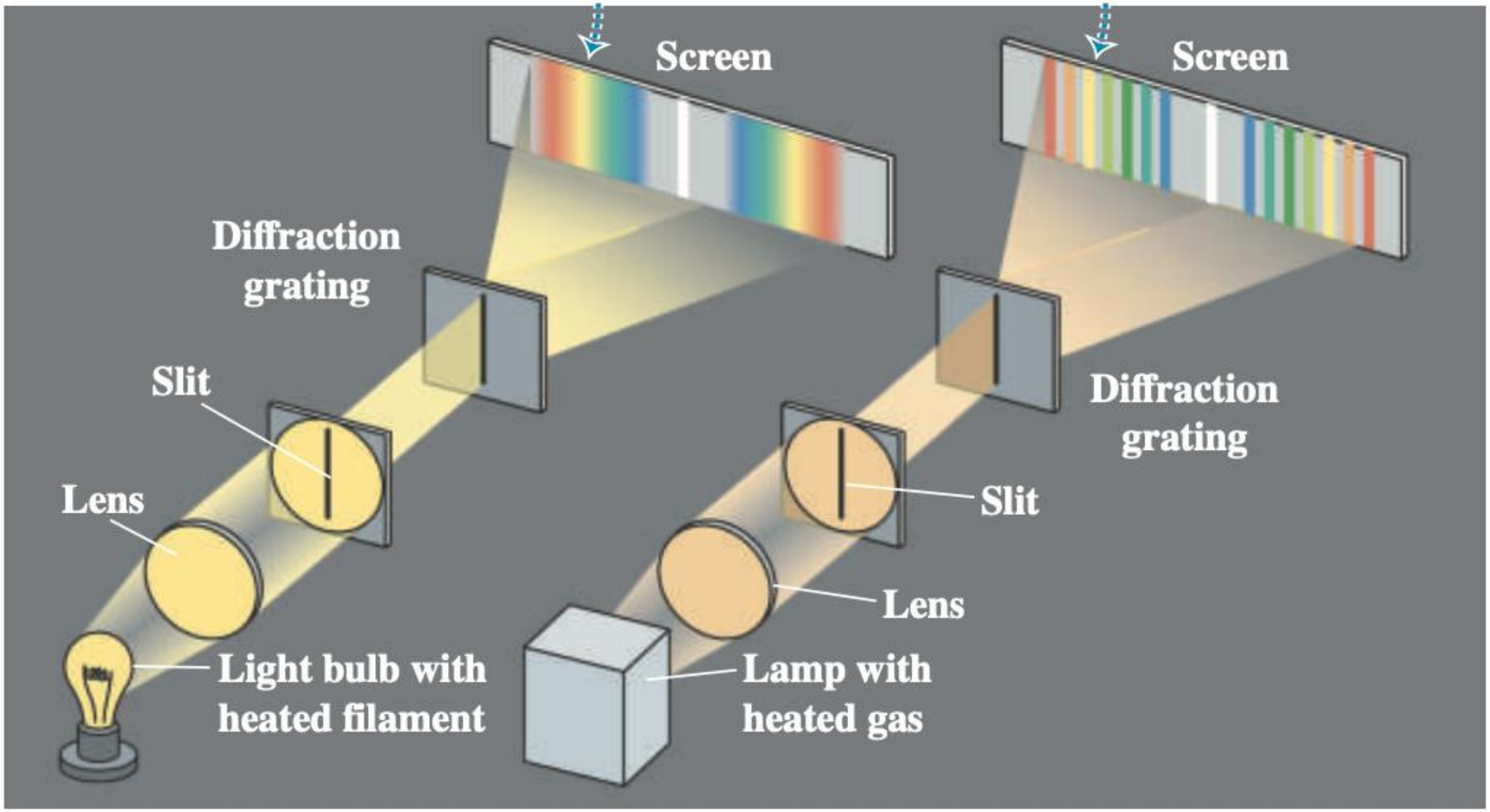
ii. Atoms should radiate energy over a continuous range of frequencies.



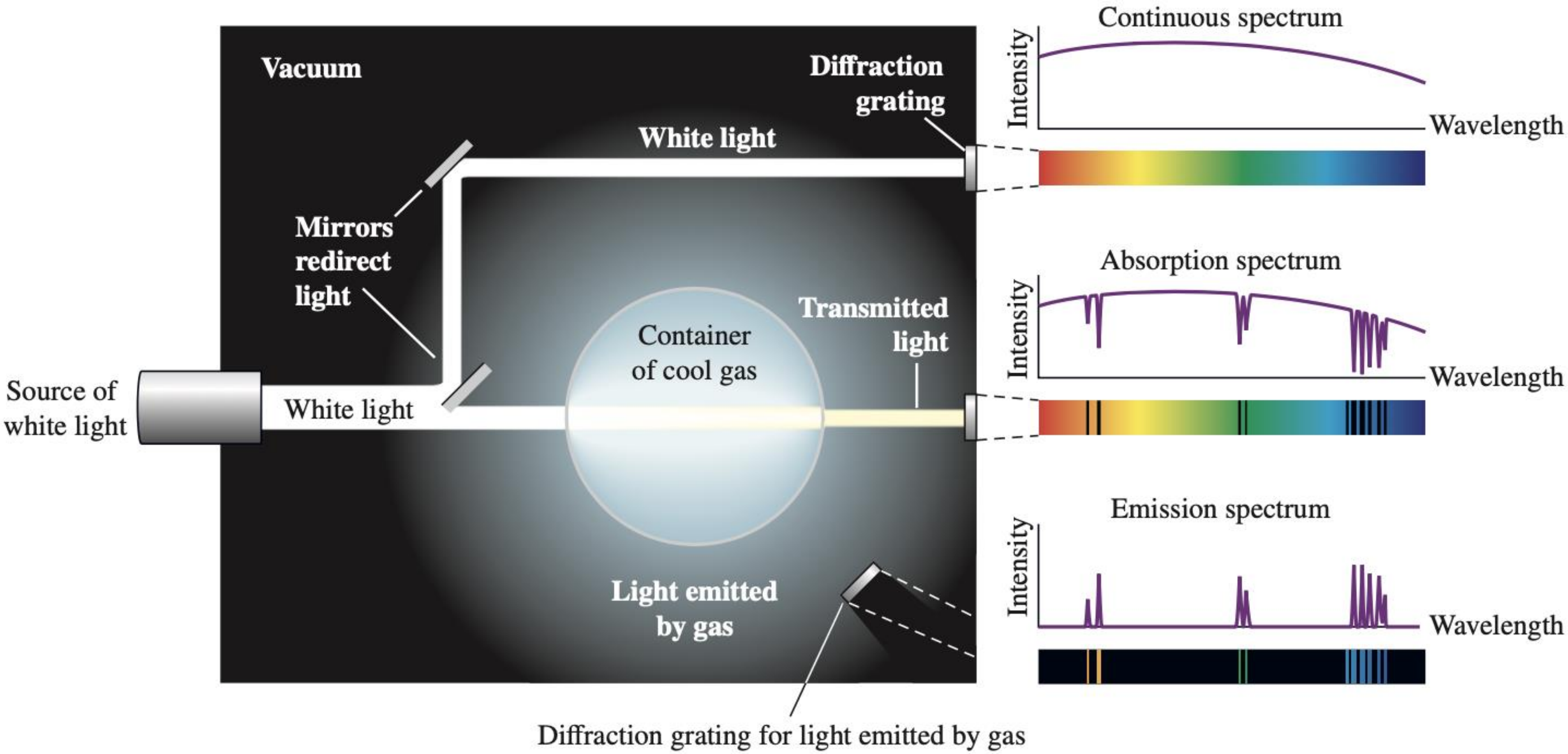
Continuous vs. Line Spectra

(a) Continuous spectrum: light of all wavelengths is present.

(b) Line spectrum: only certain discrete wavelengths are present.

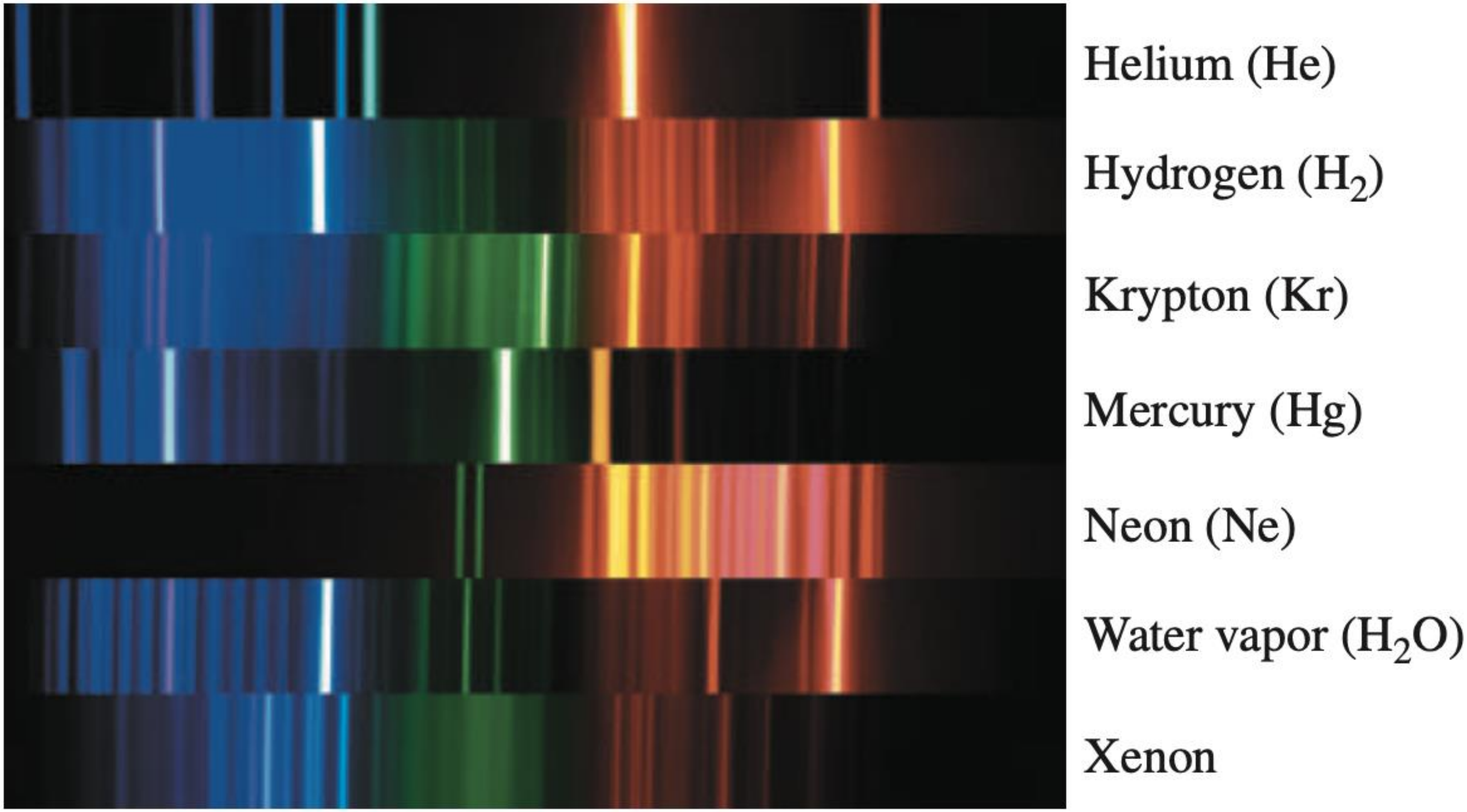


Emission vs. Absorption



Emission lines do not necessarily match absorption lines

Emission Lines are Fingerprints



Bohr's Model of Hydrogen

In 1913, Niels Bohr proposed a model of hydrogen that gave an accurate account of the observed spectrum and a convincing argument for its stability.

- Only a discrete set of stable circular orbits are allowed, and each has definite energy.
- The angular momentum in each orbit is an integer multiple of $\hbar$.
- Emission or absorption of radiation occurs when an electron jumps from one orbit to another.
  - Energy of radiation must correspond exactly to the *energy difference* between orbits.

Example 8: Bohr's Model

Apply Bohr's "quantization rules" to Rutherford's planetary model of the hydrogen atom to find

(a) the radius of the electron's circular motion for each stable orbit,

(b) the energy of the atom for each stable orbit,

(c) and the allowed wavelengths of radiation emitted by the atom.

Wave Model of Matter

In 1923, Louis de Broglie suggested that wave-particle duality was universal, and that material particles should also have a literal wave nature.

- Hypothesized that any material object with nonzero rest mass has energy and momentum given by

$$E = \hbar\omega \qquad\qquad p = |\vec{\mathbf{p}}| = \hbar k$$

$$\hbar = \frac{h}{2\pi} \qquad\qquad \omega = 2\pi f \qquad\qquad k = \frac{2\pi}{\lambda}$$

reduced Planck's constant

The *de Broglie relations* connect the energy and momentum of a particle with the frequency and wavelength of the corresponding "matter wave."

Note: this is NOT how we think of "wave functions" today!

Example 5: de Broglie Wavelength

Calculate the de Broglie wavelength for

(a) a proton of kinetic energy 70 MeV and

(b) a 100 g bullet moving at 900 m/s.

Recall the relation for relativistic momentum:

$$pc = \sqrt{K(K + 2mc^2)}$$

Example 6: Electron Gun

The de Broglie relations imply that electrons with the same momentum will have the same wavelength.

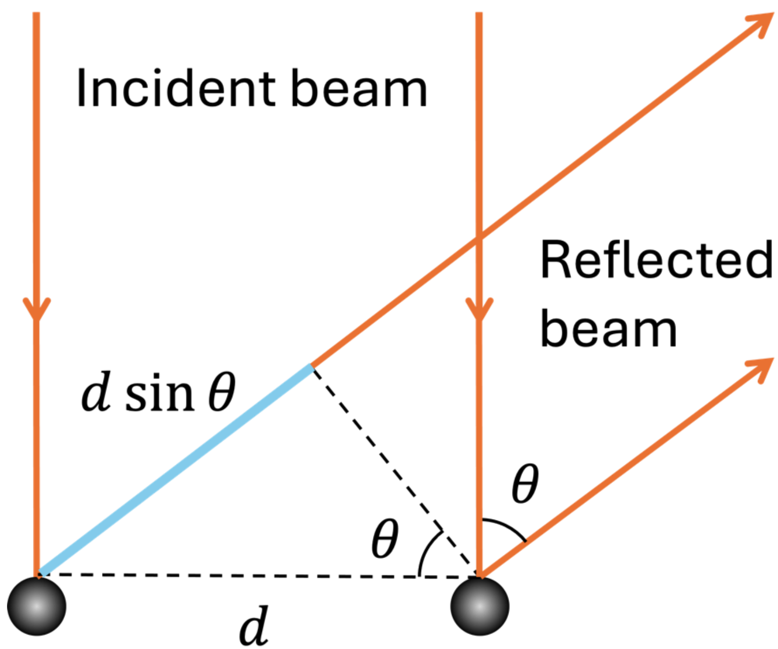
An **electron gun** is a device that accelerates electrons from rest through a fixed potential difference, thereby creating a "monochromatic beam" of electrons.

(a) Find the wavelength of electrons accelerated by a potential difference of 100 V.

(b) Why can we use the non-relativistic relation $p = \sqrt{2mK}$ in part (a)?

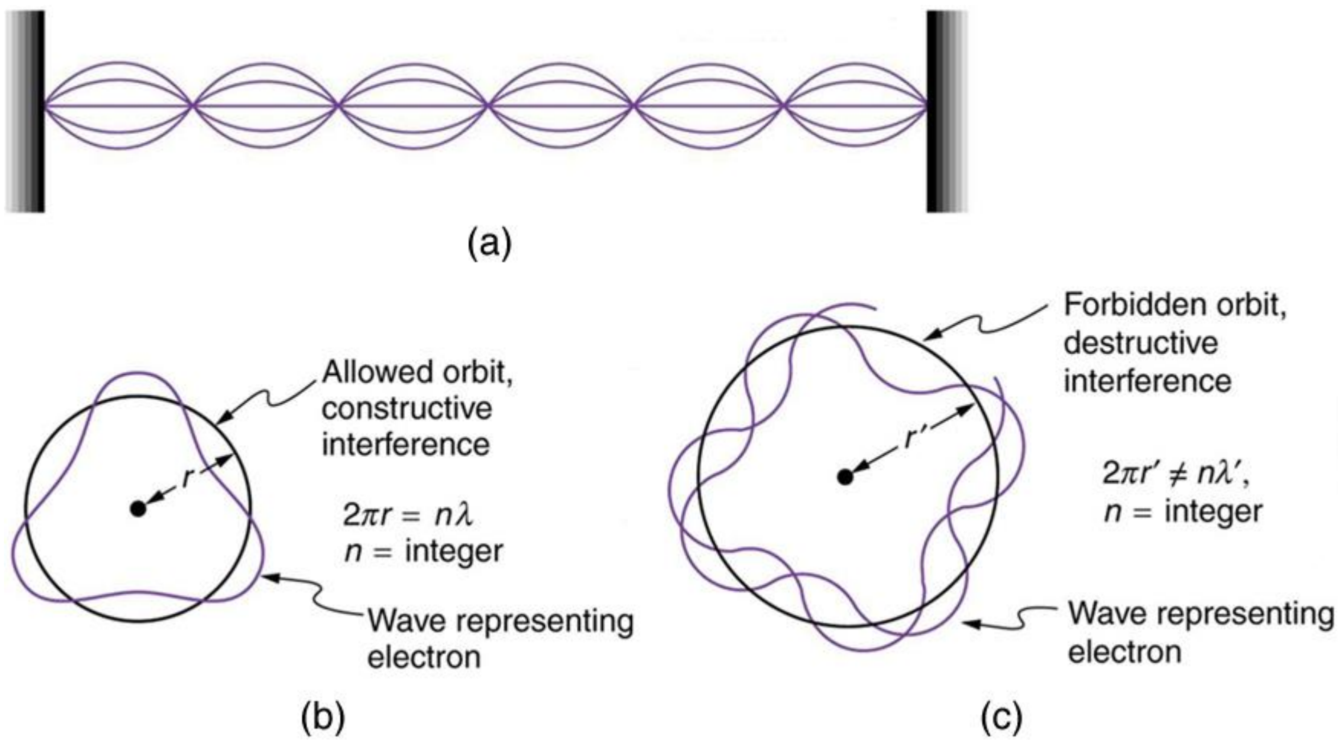
Example 7: Davisson-Germer Exp.

Electrons with a kinetic energy of 54 eV are fired toward a crystal perpendicular to its face. A detector is mounted to receive electrons reflected at a variable angle θ from the direction of the incident beam. Given that the spacing between atomic rows in nickel is $d = 0.215$ nm, what is the smallest nonzero angle for constructive interference?



Circular Standing Waves

In the Bohr model, electrons in circular orbits are treated as *standing waves*, which means the electron's de Broglie wavelength must fit an integer number of times into the circumference of the orbit.



Modern Quantum Mechanics

1. In the Spring of 1925, Werner Heisenberg developed matrix mechanics and ushers the first paradigm shift of quantum theory.
   - “The aim is to establish a formal connection between quantities which in principle are **observable**.”
   - Stop focusing on what the electron is “really doing” and instead look at the frequencies and intensities of light coming from the atom (for example).
2. In 1926, Erwin Schrödinger invented wave mechanics as an intuitive and visualizable approach.
   - Max Born points out that the wave function cannot describe a physical wave, but rather a “probability amplitude.”
   - The emphasis on **probability** was the second paradigm shift of quantum theory.
3. In 1928, Paul Dirac merged matrix and wave mechanics into a single, abstract framework—quantum mechanics.
   - A comprehensive set of postulates emerge that cover all known quantum phenomena!

1927 Solvay Conference